

OFDM

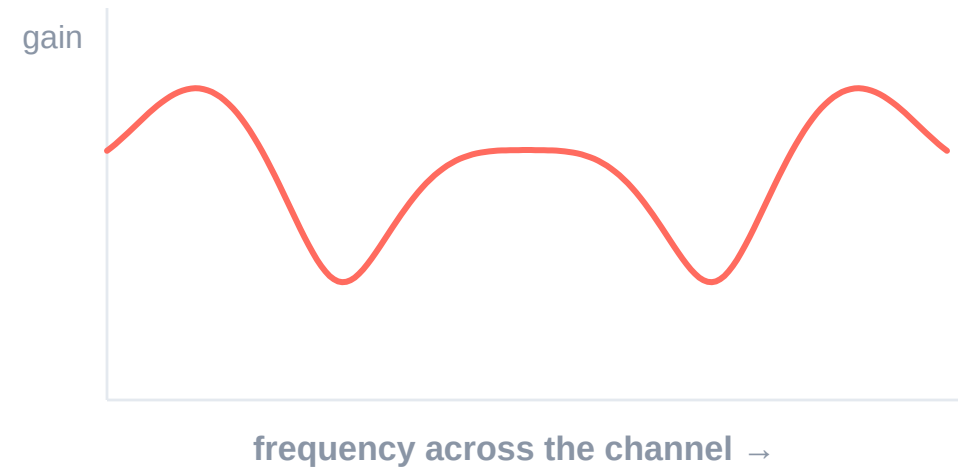
One wide channel becomes many narrow ones

A radio signal reaches the receiver several times over, along paths of different lengths. That single fact sets a limit on how fast a wideband link can run, and it is the reason OFDM exists.

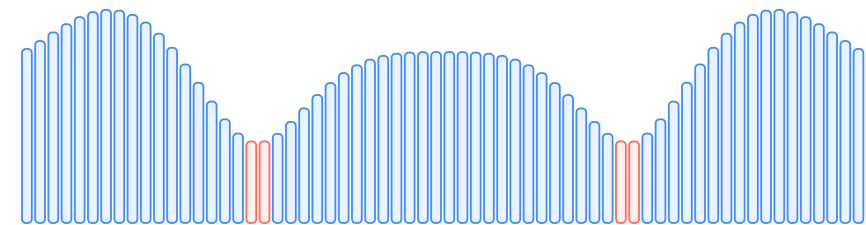
Yaxiong Xie

ONE 20 MHz CHANNEL

the gain a radio sees, frequency by frequency



SPLIT IT INTO 64 NARROW PIECES



each piece has one gain, and some are almost zero

Why do we need OFDM?

01

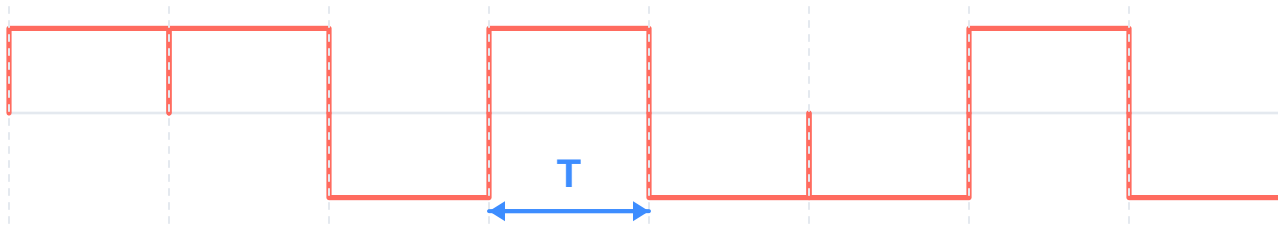
Why OFDM

A wider channel is the only way to send more symbols per second, and more symbols per second means shorter symbols. Short symbols are exactly what a multipath channel destroys. OFDM is the way out of that bind.

- Two ways to go faster

A link's rate is a symbol rate times a symbol size

$$\text{data rate} = \text{symbols per second} \times \text{bits per symbol}$$



one symbol lasts T seconds, so the link sends $1/T$ of them a second

Parts 1 and 3 are about the first factor only.

The second one comes back in the MIMO lecture.

bits per symbol

set by the signal-to-noise ratio, and it grows only with the logarithm of power

symbols per second

set by how much bandwidth you have

To raise the symbol rate you have to make each symbol shorter.

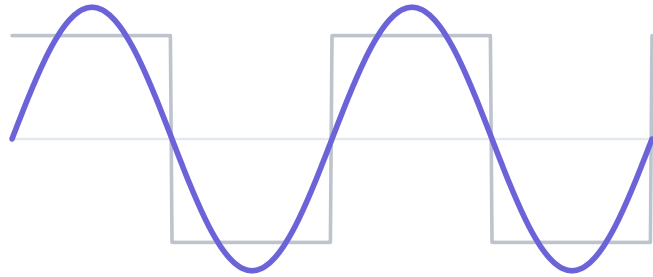
1 Why OFDM

2 The channel

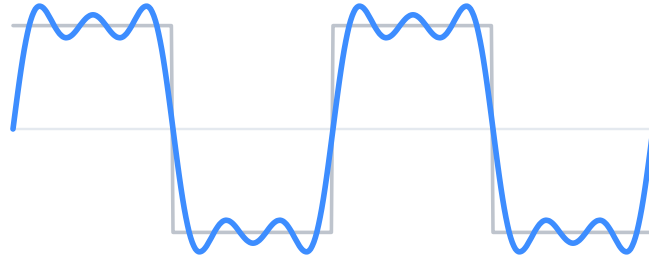
3 OFDM

A signal that changes fast contains high frequencies

BUILD A SQUARE SYMBOL STREAM OUT OF SINE WAVES



1 sine wave



3 sine waves



11 sine waves

grey is the stream we want to send; colour is what those sine waves actually add up to

The sharper the edge, the higher the frequencies you need.

Shorter symbols mean sharper edges, so they need a wider band.

Building those switches out of sine waves needs frequencies up to roughly $1/T$, and the sharper the switch the more of them.

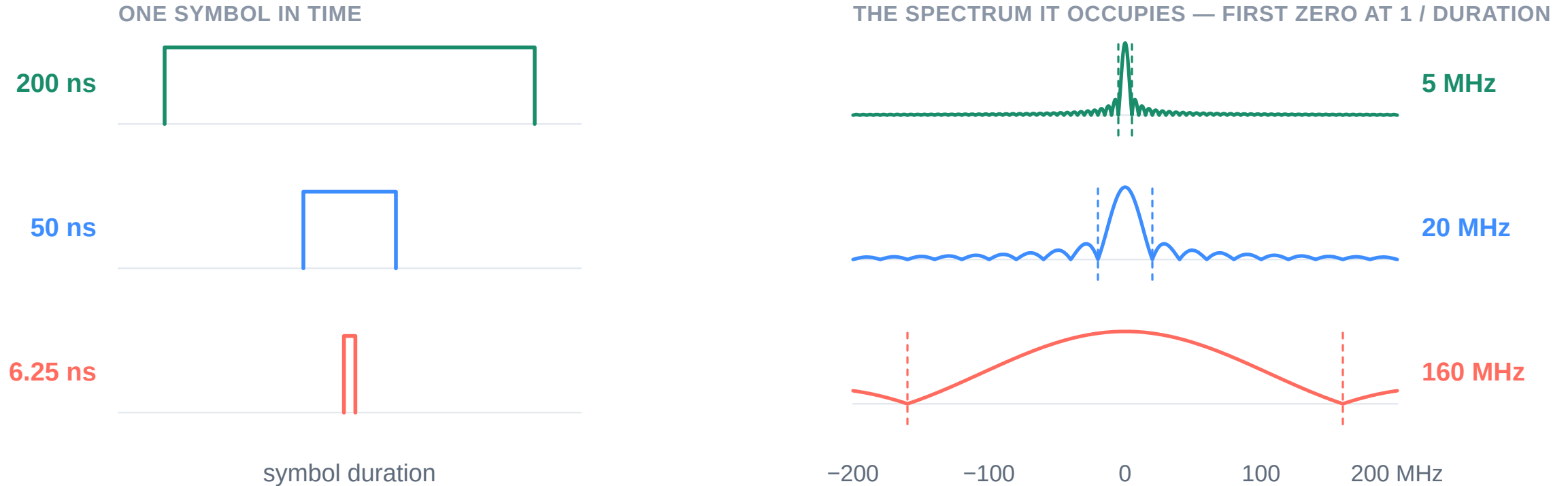
How fast the signal changes decides how much bandwidth it uses.

1 Why OFDM

2 The channel

3 OFDM

A symbol of duration T occupies about $1 / T$ hertz



The 6.25 ns symbol of a 160 MHz link spills right across the band; the 200 ns symbol of a narrow one barely moves.

Cut the symbol to a quarter of its length and the spectrum is four times wider.

1 Why OFDM

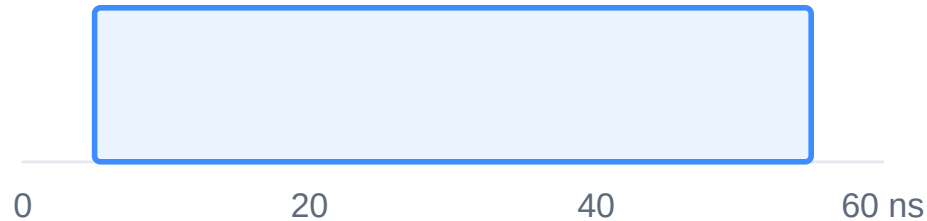
2 The channel

3 OFDM

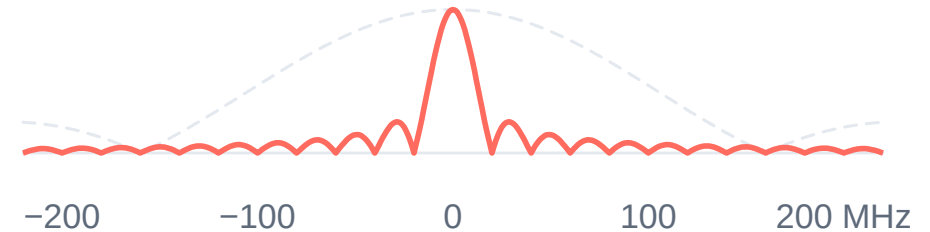
Bandwidth and symbol rate are the same number

$$\text{symbol time } T \approx 1 / \text{bandwidth}$$

ONE SYMBOL IN TIME



THE SPECTRUM IT OCCUPIES

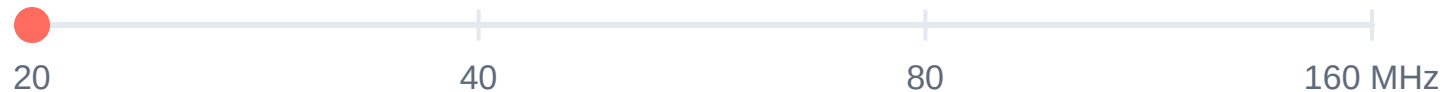


CHANNEL WIDTH

20 MHz

SYMBOL TIME

50 ns



The animation sweeps the channel width from 20 to 160 MHz. The symbol shrinks and its spectrum widens by the same factor, because the two are locked together.

Widening the channel and shortening the symbol are one action.

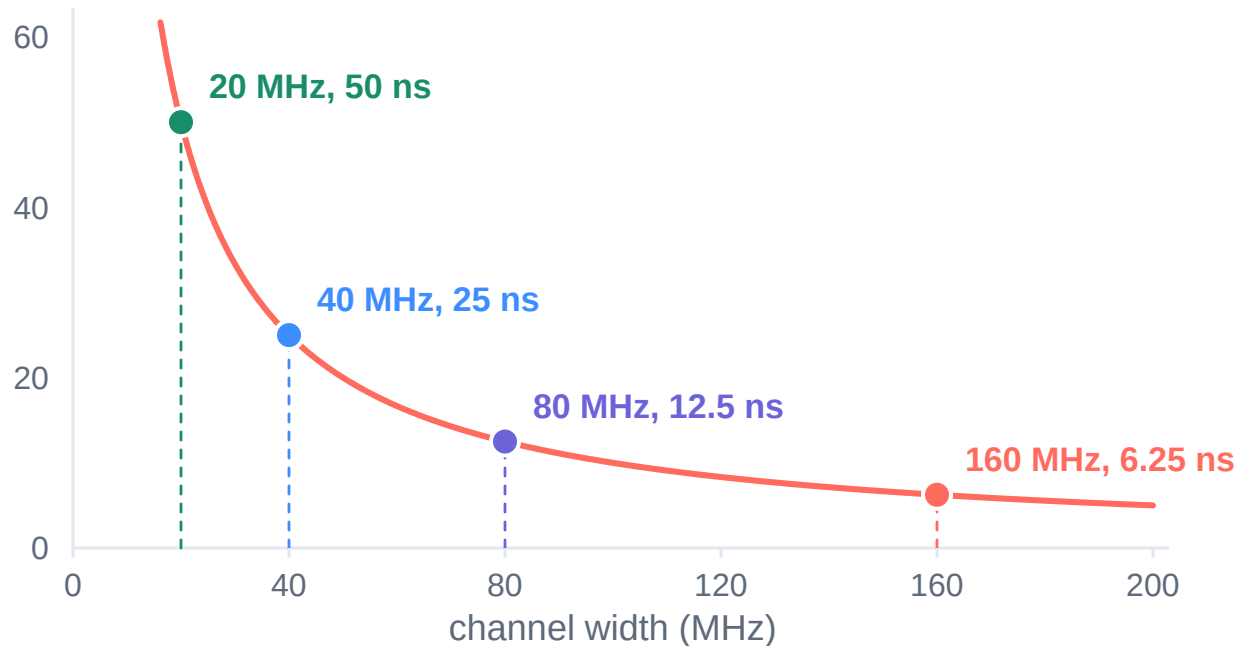
1 Why OFDM

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Symbol time falls as one over the bandwidth

symbol time (ns)



WIDTH	SYMBOLS / s	SYMBOL TIME
20 MHz	20 M	50 ns
40 MHz	40 M	25 ns
80 MHz	80 M	12.5 ns
160 MHz	160 M	6.25 ns

Double the width, double the symbol rate, halve the symbol time.

20 to 40 MHz saves 25 ns of symbol time. 160 to 180 MHz saves 0.7 ns.

Every doubling of the channel halves the symbol.

1 Why OFDM

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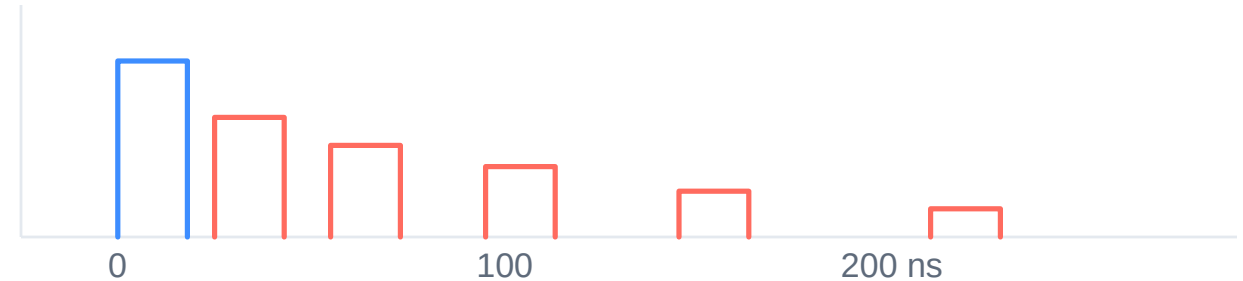
One transmitted symbol arrives many times

WHAT THE TRANSMITTER SENDS



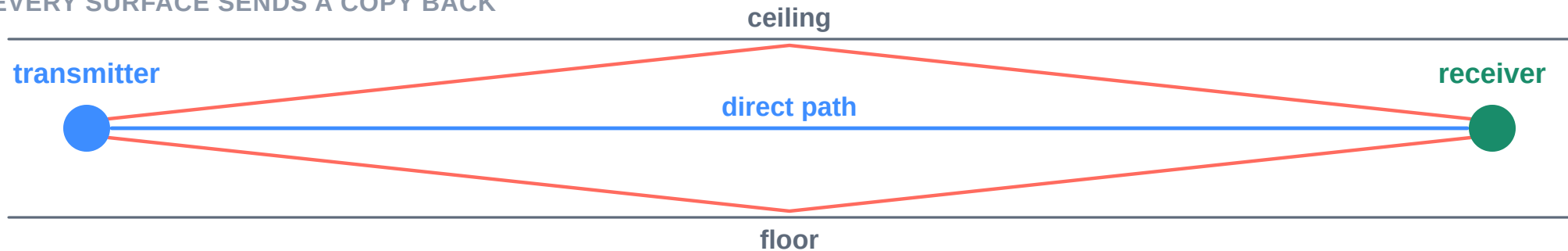
one symbol

WHAT THE RECEIVER GETS



the same symbol, six times, at six delays

WHY: EVERY SURFACE SENDS A COPY BACK



Each reflected path travels further, so it arrives later and weaker. The receiver adds all of them together; it cannot separate them.

The channel is a sum of delayed, scaled copies.

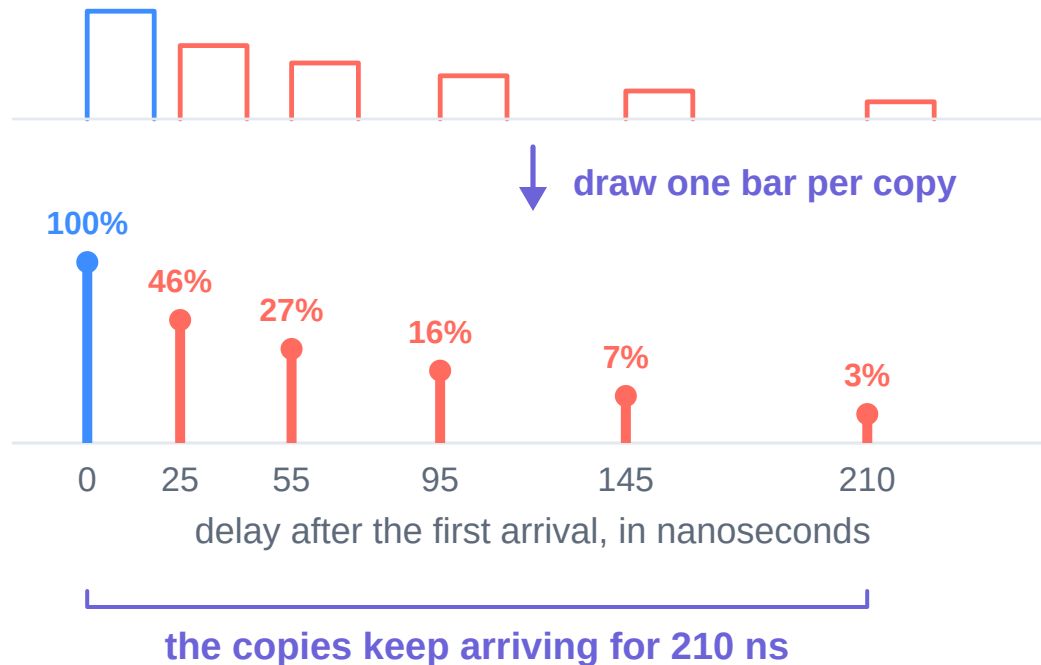
1 Why OFDM

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Delay spread is how long the copies keep arriving

THE COPIES FROM THE LAST SLIDE, ON A TIME AXIS



DELAY SPREAD

how long the copies keep arriving

here: 210 ns from the first to the last

RMS DELAY SPREAD

the same span, averaged and weighted by how strong each copy is

here: 42 ns

this is the number measurement tables quote

The same six copies, each placed at its delay and drawn as tall as its strength. Engineers call this the power delay profile.

One number summarises the channel: how far apart in time the copies land.

1 Why OFDM

2 The channel

3 OFDM

The delays are tens to thousands of nanoseconds

60 ns

median RMS delay spread in an office at 5 GHz

18 m of extra path length

Light travels 30 cm in 1 ns, so 60 ns of spread
is an 18 m difference in path length.

The copies themselves keep arriving five to ten
times longer than the RMS number.

Indoor values are ITU-R P.1238 measurements at 5.2 GHz. The outdoor value is the ITU-R M.1225 Vehicular B model, whose last copy arrives 20 μ s late.

RMS DELAY SPREAD, MEDIAN

home	23 ns
office	60 ns
shopping mall	190 ns
street, moving car	4,000 ns

Delay spread is a property of the room, not of the radio.

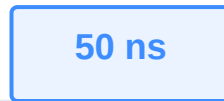
1 Why OFDM

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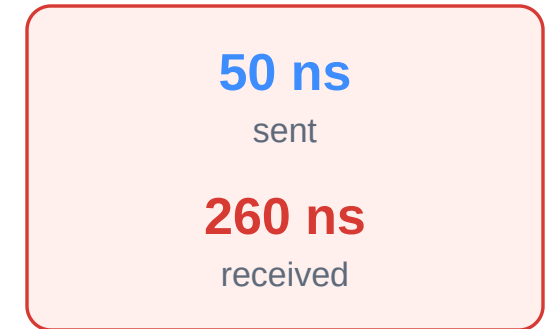
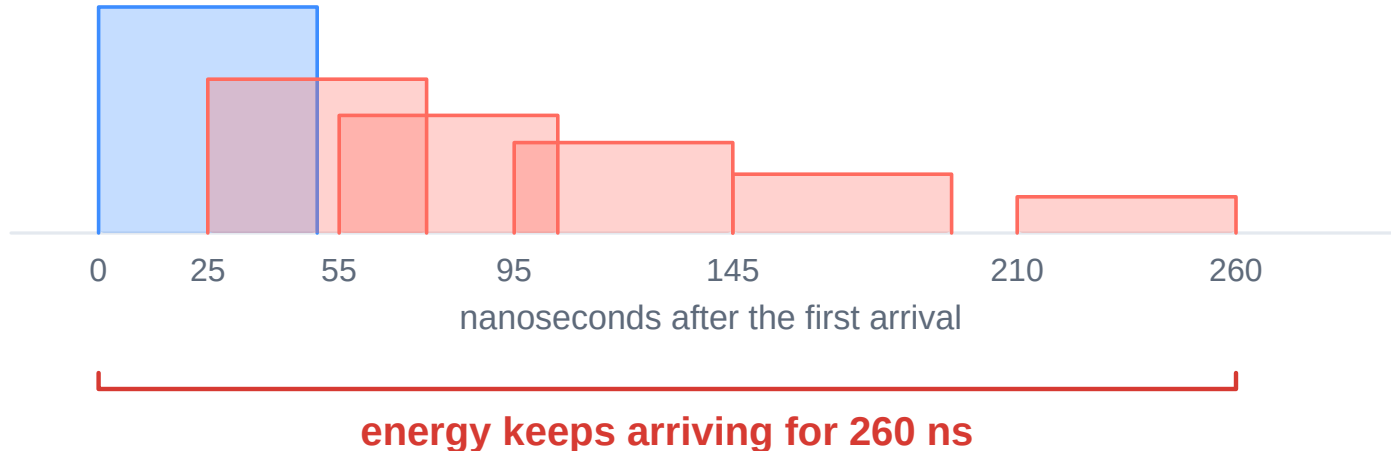
3 OFDM

A symbol comes back longer than it was sent

SENT: ONE 50 ns SYMBOL



RECEIVED: SIX COPIES, EACH THE WHOLE SYMBOL, ONE ON TOP OF ANOTHER



50 + 210
symbol plus delay spread

Nothing was added to the symbol. The same 50 ns of signal arrives six times over, spread across 210 ns, so the receiver sees energy from it for 260 ns.

What arrives lasts the symbol plus the delay spread.

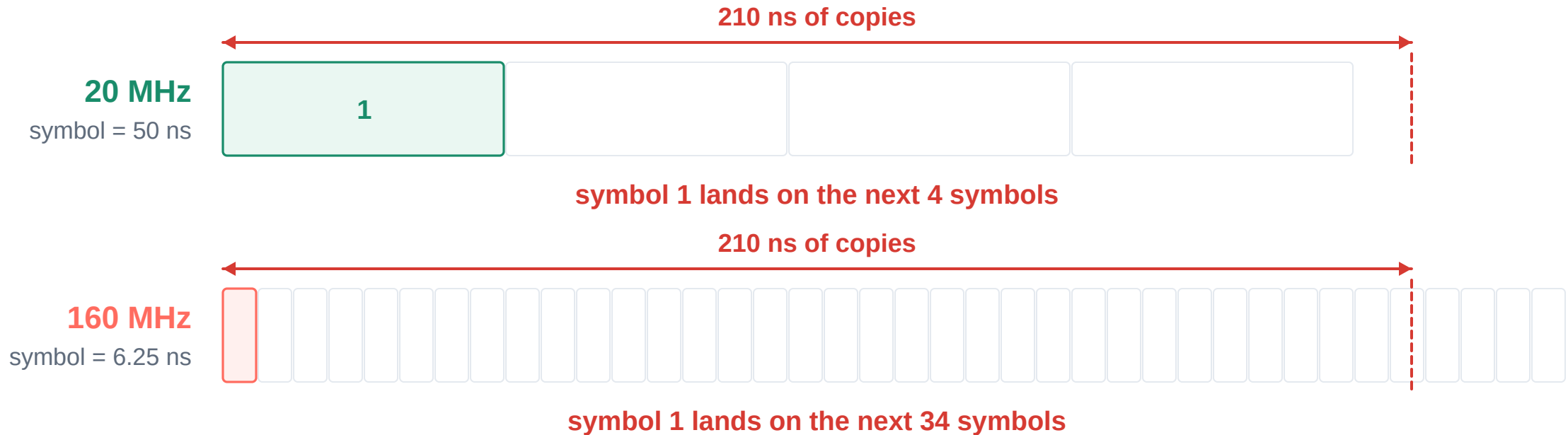
1 Why OFDM

2 The channel

3 OFDM

A wider channel puts more symbols inside the same 210 ns

THE FIRST 240 ns OF A TRANSMISSION, AT TWO CHANNEL WIDTHS



The channel did not change. The symbols got shorter, so more of them fall inside the same 210 ns.

Symbol 1 is still arriving while the symbols after it are being sent. Each one lands on top of the others. That is intersymbol interference.

The wider the channel, the more symbols overlap.

1 Why OFDM

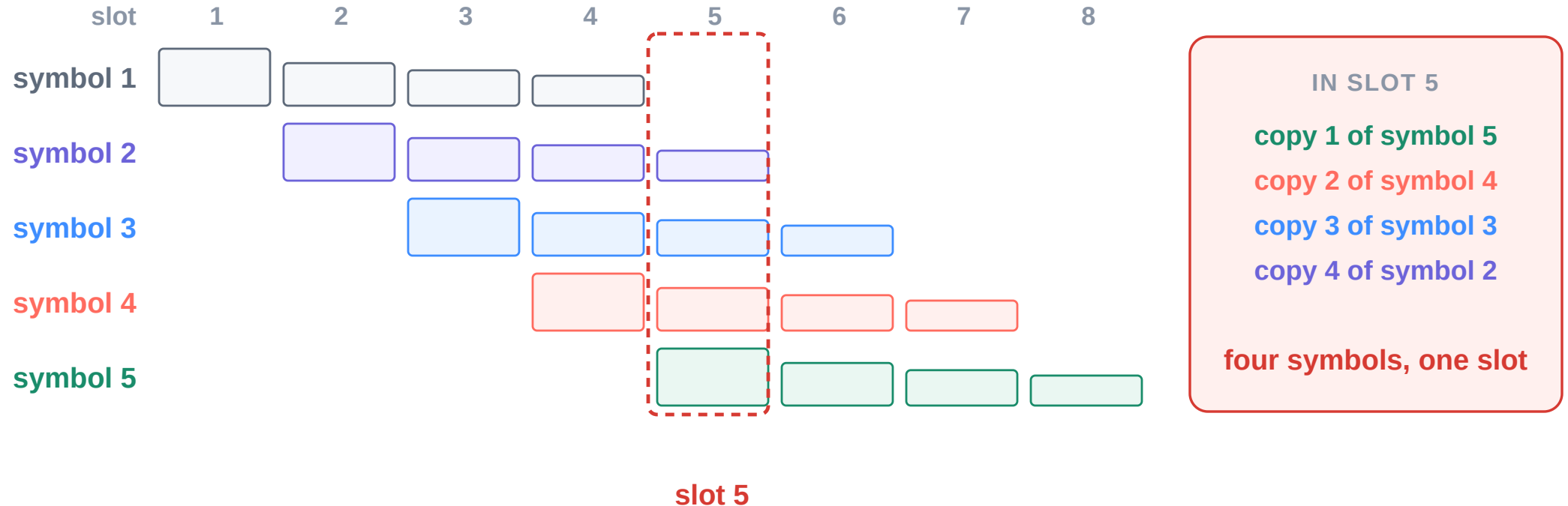
2 The channel

3 OFDM

- — One symbol, four slots

Each symbol spreads forward over the slots after it

EVERY SYMBOL ARRIVES FOUR TIMES: IN ITS OWN SLOT AND THE THREE AFTER IT



One row per transmitted symbol: its own slot at full strength, then the next three at 68%, 52% and 40%.

Symbol 5 has three older symbols sitting on top of it.

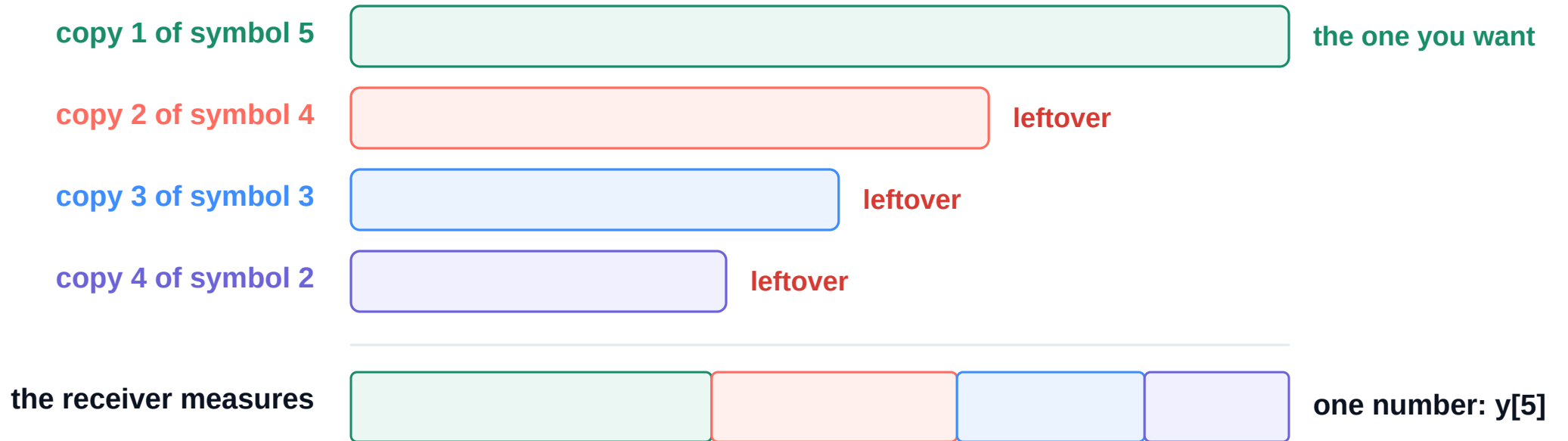
1 Why OFDM

2 The channel

3 OFDM

Four things land in the same slot and only one is wanted

THE FOUR THINGS THAT LAND IN SLOT 5, DRAWN BY STRENGTH



The four contributions arrive on top of each other. The receiver cannot see them separately.

To read symbol 5, the three leftovers have to come off first.

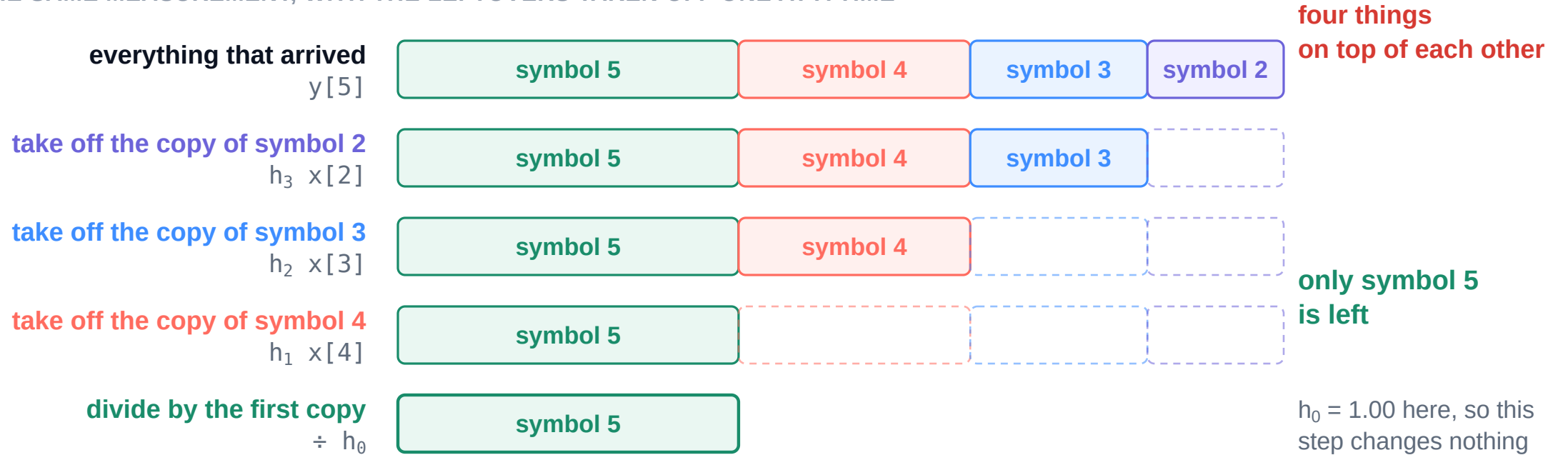
1 Why OFDM

2 The channel

3 OFDM

Take the leftovers off, one copy at a time

THE SAME MEASUREMENT, WITH THE LEFTOVERS TAKEN OFF ONE AT A TIME



The receiver already decided symbols 2, 3 and 4 in the slots before this one, so it knows every leftover and can take it off.

Three multiplications and three subtractions for every single symbol.

1 Why OFDM

2 The channel

3 OFDM

The number of subtractions grows with the channel width

$$\text{subtractions per symbol} = \text{delay spread} \div \text{symbol time}$$

CHANNEL	SYMBOL TIME	COPIES ARRIVE FOR	SUBTRACTIONS PER SYMBOL
20 MHz, indoors	50 ns	210 ns	 4
80 MHz, indoors	12.5 ns	210 ns	 17
160 MHz, indoors	6.25 ns	210 ns	 34
160 MHz, outdoors	6.25 ns	20,000 ns	 3,200

The work grows with the bandwidth

widening the channel is what shortens the symbol

A wrong decision poisons the next ones

every subtraction uses symbols the receiver already guessed at

At 160 MHz outdoors the receiver carries 3,200 past symbols into every new one.

1 Why OFDM

2 The channel

3 OFDM

Two things we need, and they pull against each other

WHAT WE WANT

A wide channel.

It is the only way to send more symbols per second.

WHAT IT COSTS

A short symbol.

And a short symbol is buried under copies of the ones before it.

**Wide channel and long symbol
look like opposites.**

The rest of Part 1 is one question: can a signal be wide in frequency and slow in time at once?

Everything so far says no. The next slide says yes.

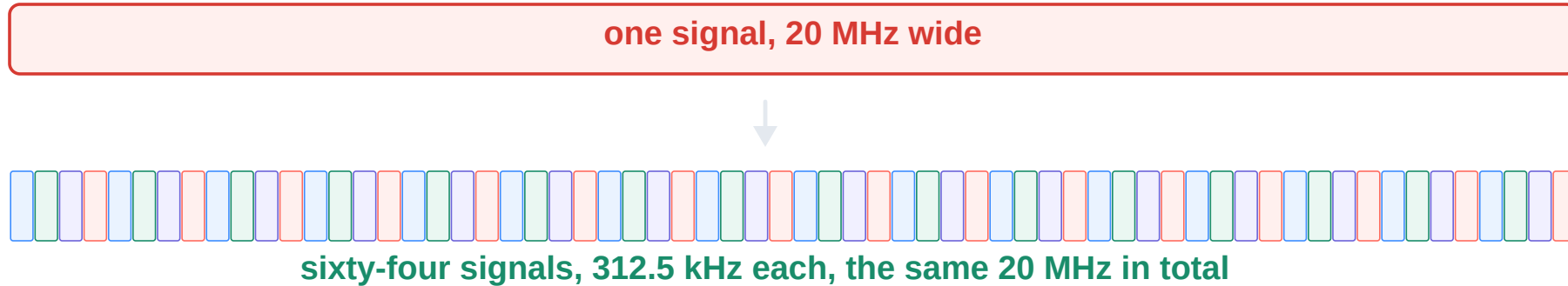
1 Why OFDM

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3 OFDM

Split the wide signal into many narrow ones

IN FREQUENCY — ONE WIDE SIGNAL, OR MANY NARROW ONES



IN TIME — ONE SYMBOL OF EACH, DRAWN TO THE SAME SCALE



time, 0 to 3.2 microseconds

Slide 6 said a narrow signal is a slow one. Cutting the band into 64 pieces makes each piece 64 times narrower, and therefore each symbol 64 times longer.

Same total bandwidth, symbols sixty-four times longer.

1 Why OFDM

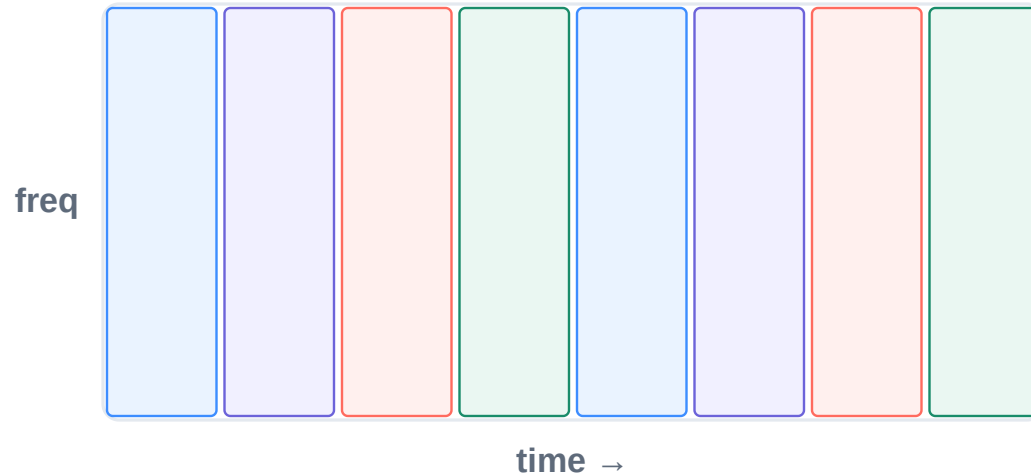
2 The channel

3 OFDM

- — The same box, tiled differently

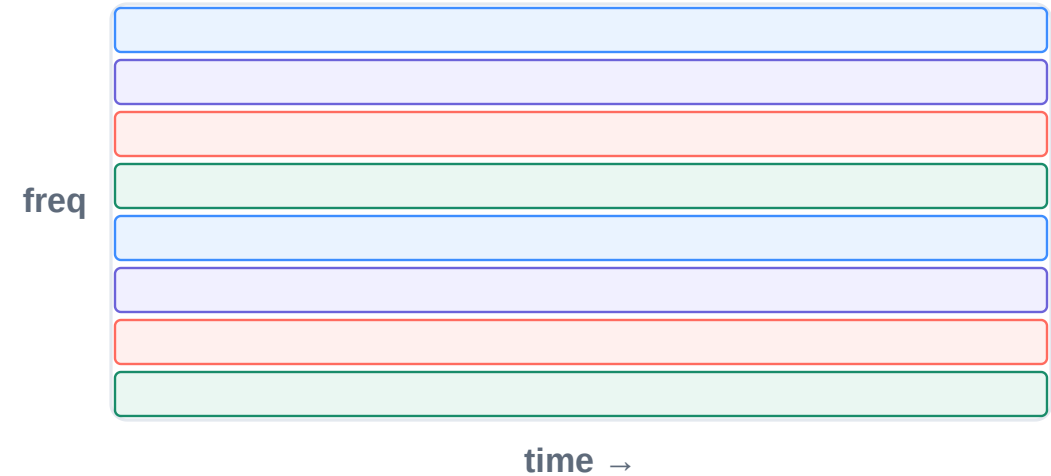
Many symbols in time, or many in frequency

BEFORE: ONE WIDE SIGNAL



one symbol at a time, each 50 ns long
many symbols, one after another in time

OFDM: MANY NARROW SIGNALS



eight symbols at once, each 3.2 μ s long
many symbols, side by side in frequency

Both rectangles are the same size: same 20 MHz, same 3.2 μ s, same eight symbols. What changed is the shape of one symbol.

Drawn with eight pieces instead of sixty-four so the tiles are visible.

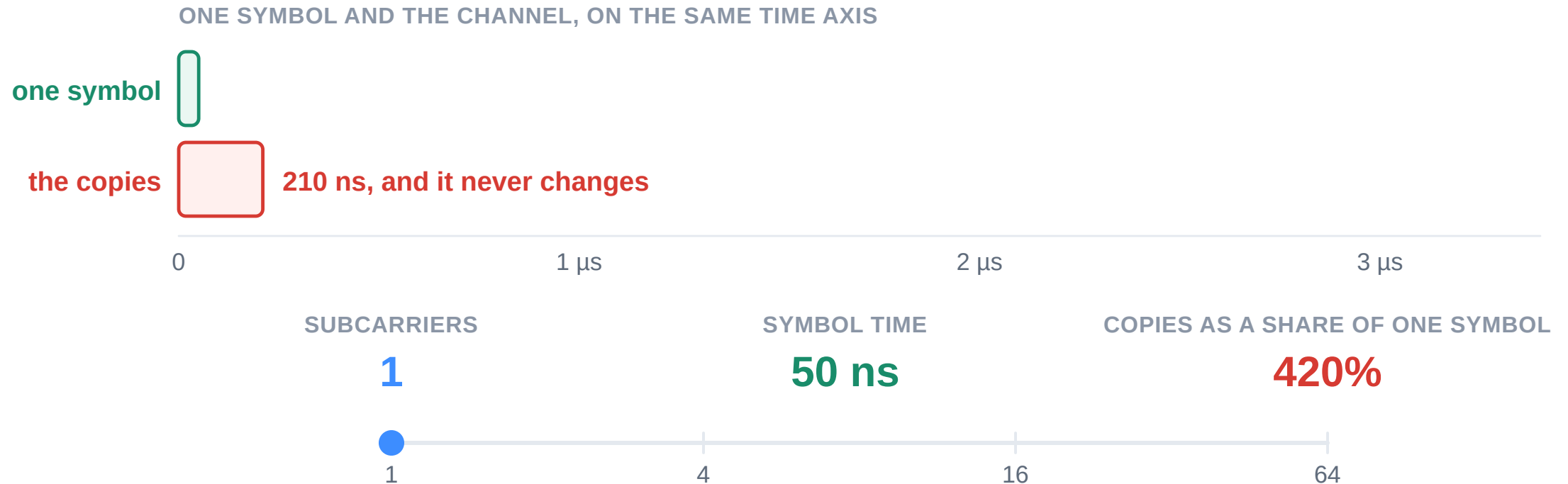
OFDM does not send more symbols. It sends them in a different shape.

1 Why OFDM

2 The channel

3 OFDM

Split the band and the same copies stop mattering



The red bar is the channel and never moves. The green bar is one symbol, and it grows as the band is split into more pieces.

The channel did not change. The symbol got longer.

1 Why OFDM

2 The channel

3 OFDM

What exactly does multipath do to a signal?

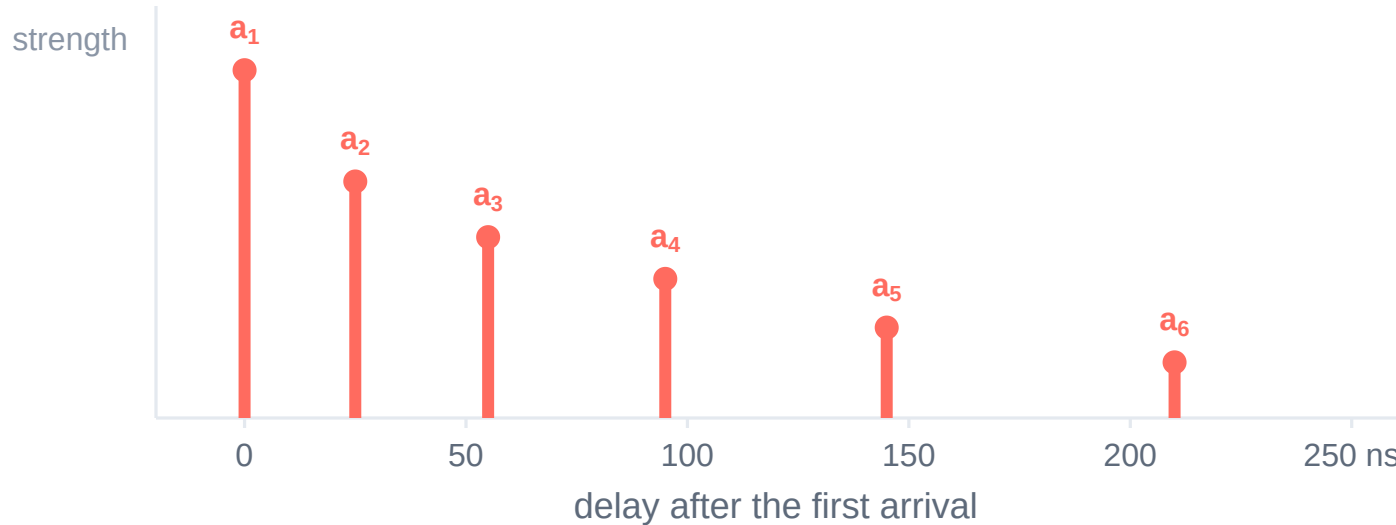
02

The channel

The same physical fact has two descriptions. In time it is a list of delayed copies. In frequency it is a gain that changes across the channel. The second description is the one OFDM is built on.

The channel is a list of delays and gains

THE CHANNEL IMPULSE RESPONSE $h(\tau)$



$$h(\tau) = \sum_i a_i \delta(\tau - \tau_i)$$

one tap

one path: a delay and a gain

τ_{rms}

the spread of the delays, power weighted

τ_{max}

the delay of the last usable tap

For this channel $\tau_{\text{rms}} = 42$ ns and $\tau_{\text{max}} = 210$ ns.

The receiver never sees the taps one by one. It sees all six copies added together, as one signal.

Each tap is one bundle of paths that arrive at about the same time. A tap gain is complex: the path changes both the amplitude and the phase.

So the question is how a signal and a delayed copy of it add up.

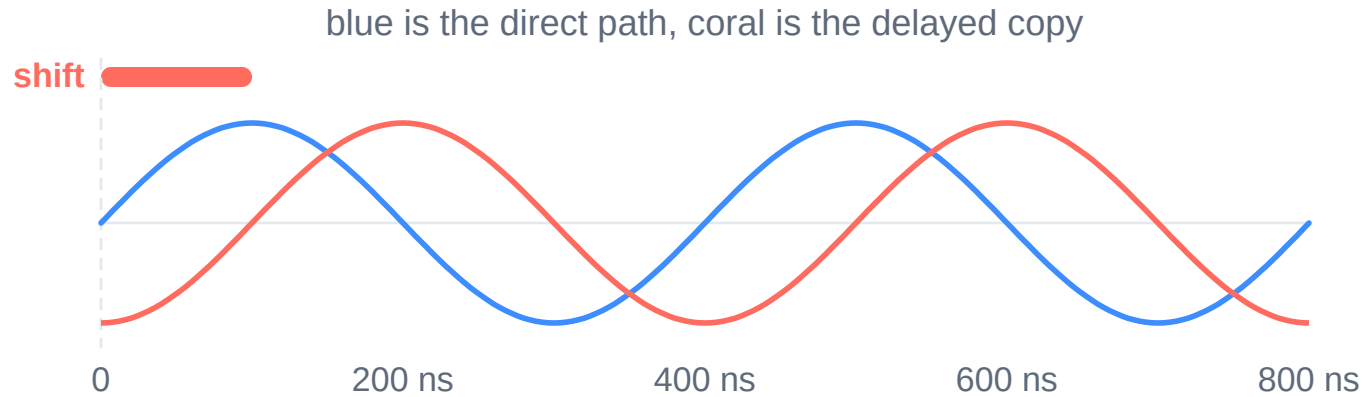
1 Why OFDM

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3 OFDM

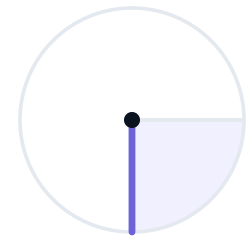
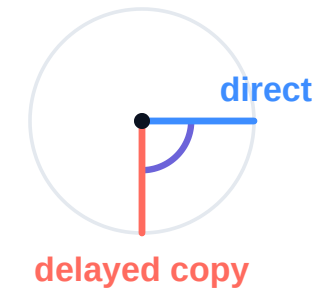
A delay slides the wave sideways, and nothing else

ONE TONE AT 2.5 MHz — ONE CYCLE LASTS 400 ns



EACH PATH

THE DIFFERENCE



SHIFT IN TIME

100 ns

EXTRA PATH LENGTH

30 m

AS A FRACTION OF A CYCLE

0.25

PHASE DIFFERENCE

90°

It is the same wave, later. The angle between the two hands is set by how much later, and 30 m of extra path is 100 ns.

Phase is the shift, measured as a fraction of one cycle.

1 Why OFDM

2 The channel

3 OFDM

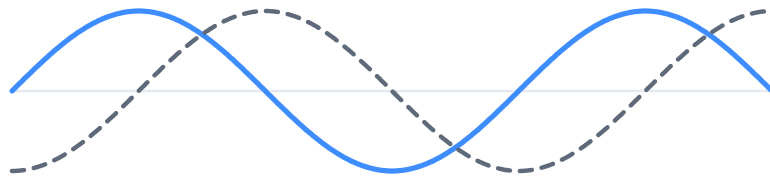
One delay, but a different phase at every frequency

$$\text{phase} = 2\pi \times f \times \tau$$

the delay τ is fixed by the path length. The phase it produces is not — it is proportional to the frequency.

tone at 2.5 MHz

one cycle lasts 400 ns



$\tau = 100$ ns, the same in both

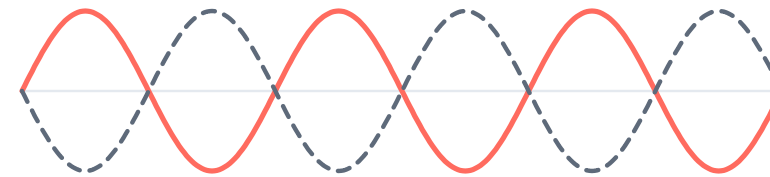


phase 90°

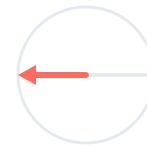
a quarter of a cycle

tone at 5 MHz

one cycle lasts 200 ns



$\tau = 100$ ns, the same in both



phase 180°

half a cycle — the copy is upside down

Solid is the direct copy, dashed is the delayed one, and the bar under each pair is the same 100 ns.

Double the frequency and the same delay costs twice the phase.

1 Why OFDM

2 The channel

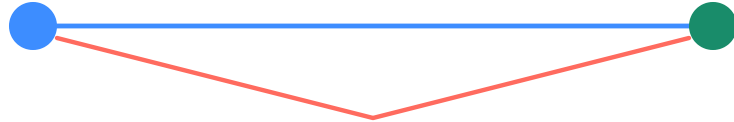
3 OFDM

Two equal paths half a cycle apart cancel exactly

A DIRECT PATH AND ONE REFLECTION, 30 m LONGER: $\tau = 100$ ns

transmitter

receiver



30 m further, so 100 ns later

Now send a 5 MHz tone into it.

One cycle lasts 200 ns, so 100 ns of delay is exactly half a cycle.

The two arrivals are upside down with respect to each other.

direct path



reflected path



they add up to



nothing

One copy is 100 ns behind, which at 5 MHz is half a cycle, so wherever one is positive the other is negative by the same amount.

A path difference of 30 m can switch off a 5 MHz tone completely.

1 Why OFDM

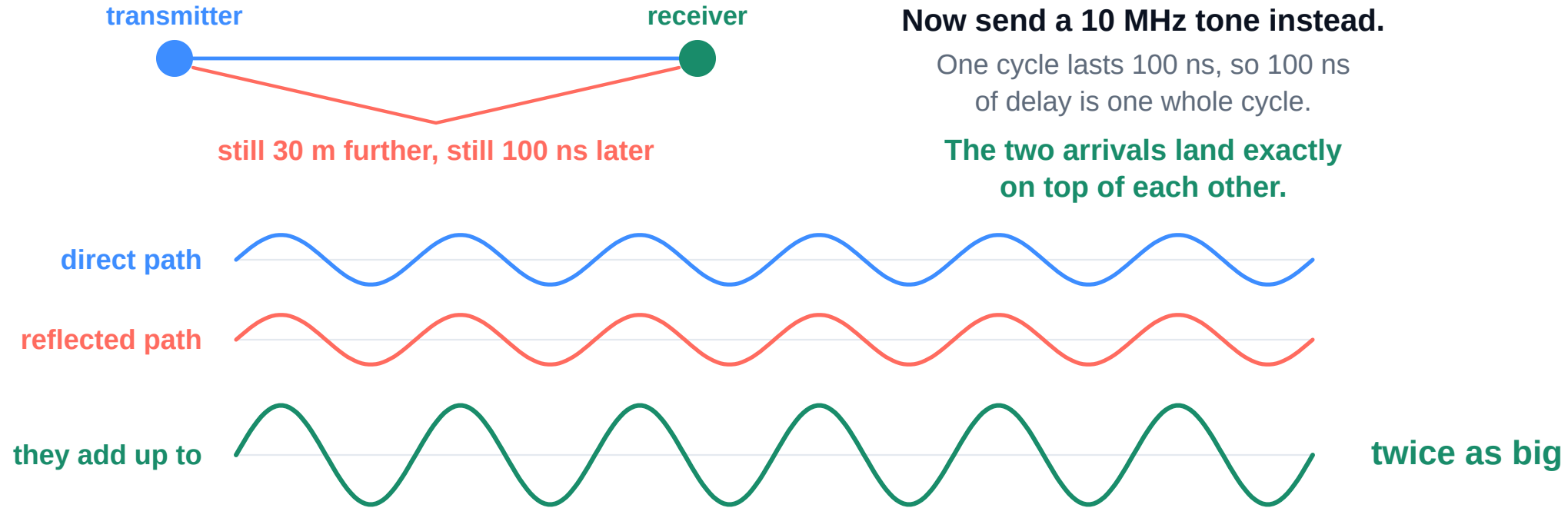
2 The channel

3 OFDM

- — The same paths, a tone that fits

Two equal paths a whole cycle apart double

THE SAME TWO PATHS. THE SAME 100 ns. A DIFFERENT TONE.



Nothing in the room changed. Only the transmitted frequency is different, and the answer went from nothing to twice one path.

The same two paths cancel one tone and reinforce another.

1 Why OFDM

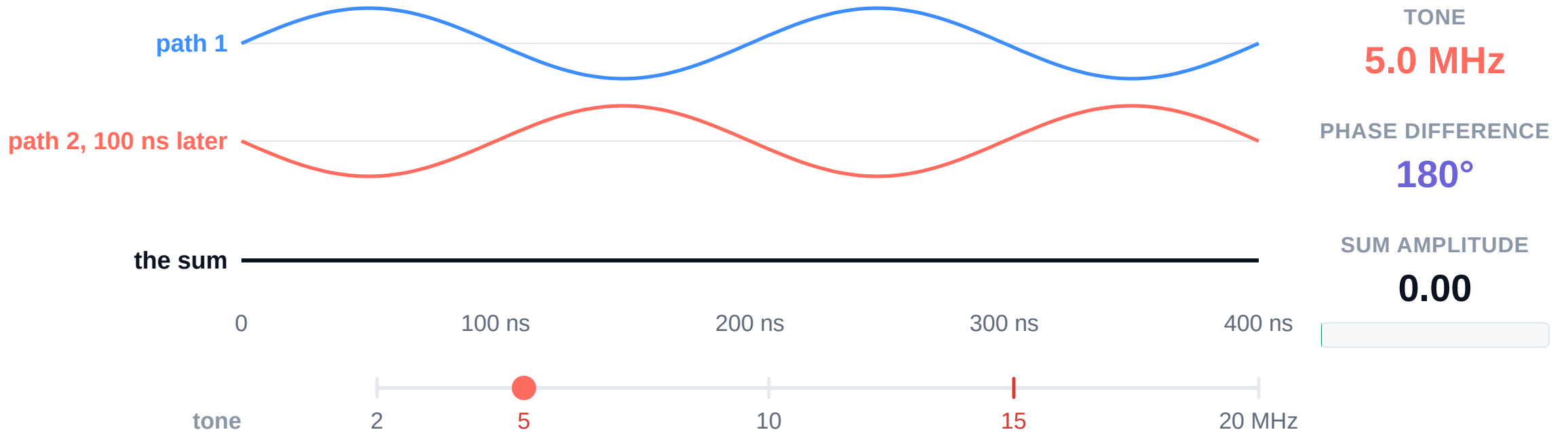
2 The channel

3 OFDM

● — Same paths, a different tone

Nothing moved in the room, and the answer changed

THE SAME TWO PATHS, $\tau = 100$ ns. ONLY THE TRANSMITTED TONE CHANGES.



The two paths never move. Only the transmitted tone is swept, and the sum runs from twice one path down to nothing and back.

The same channel helps one tone and kills another.

1 Why OFDM

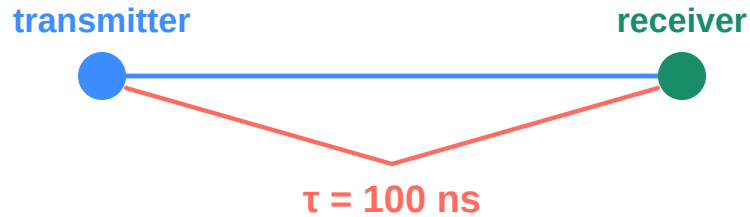
2 The channel

3 OFDM

- — Every frequency at once

One signal, and a different answer at each frequency

THE SETUP, UNCHANGED: ONE DIRECT PATH AND ONE 30 m LONGER



So far we sent one tone at a time.

A real signal is every frequency in its band, all at once. Send all of 900 to 920 MHz into these same two paths and ask what each frequency inside it gets.

THE PHASE DIFFERENCE AT FIVE FREQUENCIES INSIDE THAT BAND

FREQUENCY	CYCLES IN 100 ns	PHASE DIFFERENCE	THE TWO PATHS
900 MHz	90.0	0°	add up
905 MHz	90.5	180°	cancel
910 MHz	91.0	0°	add up
915 MHz	91.5	180°	cancel
920 MHz	92.0	0°	add up

One signal goes in. Different frequencies inside it come out with completely different gains.

The phase difference turns a full circle every $1/\tau = 10 \text{ MHz}$.

1 Why OFDM

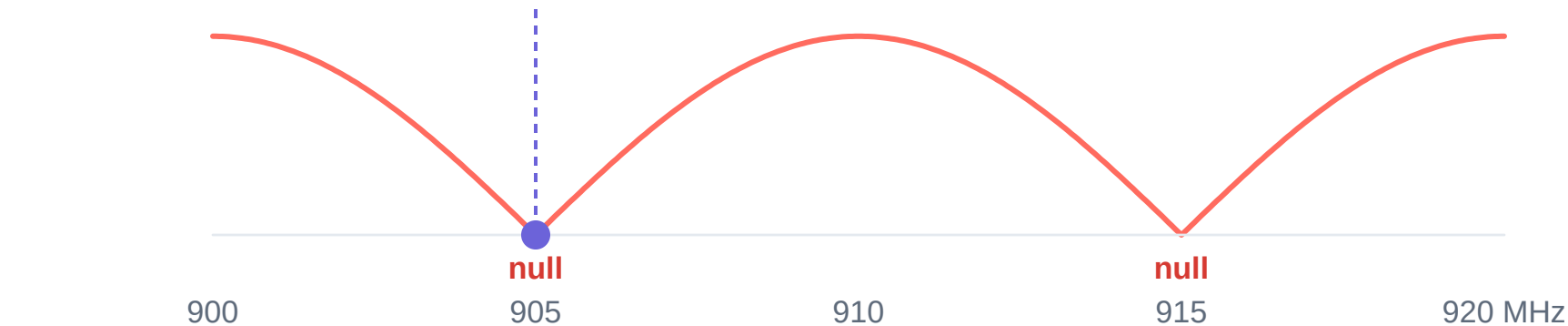
2 The channel

3 OFDM

Twenty megahertz comes back with holes in it

SENT: ALL OF 900 TO 920 MHz AT ONCE

THE CHANNEL: TWO ARRIVALS OF THAT SIGNAL, ONE 100 ns AFTER THE OTHER, ADDED AT EVERY FREQUENCY

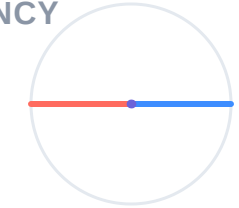


RECEIVED: THE SIGNAL WITH TWO BITES TAKEN OUT OF IT



At every frequency the marker adds the two arrivals with the phase difference that frequency gives. 100 ns is one cycle at 10 MHz, so the pattern repeats every 10 MHz.

THE TWO ARRIVALS BEING ADDED



path 1 + path 2

FREQUENCY
905.0 MHz

PHASE DIFFERENCE

180°

GAIN

0.00

One delay, one bandwidth, and the nulls sit $1/\tau$ apart.

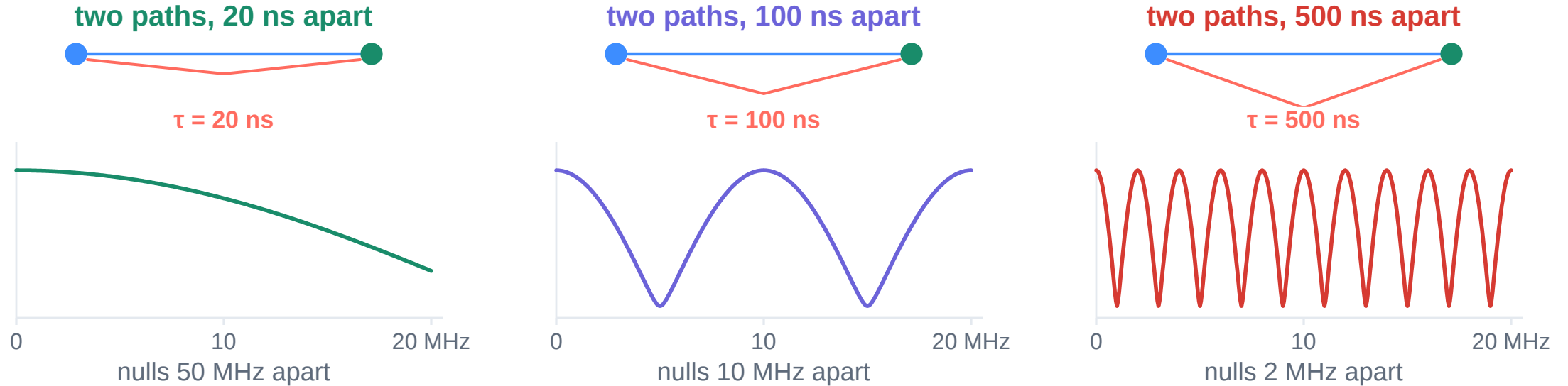
1 Why OFDM

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Delay decides how many nulls fit in one channel

IN EVERY PANEL: TWO ARRIVALS OF THE SAME SIGNAL, ADDED. ONLY THE GAP BETWEEN THEM CHANGES.



The gap between the two arrivals sets the spacing of the nulls, and that spacing is $1 / \tau$.

All three plots span the same 20 MHz. Only the gap between the two arrivals changes.

A longer gap between arrivals packs more nulls into the same channel.

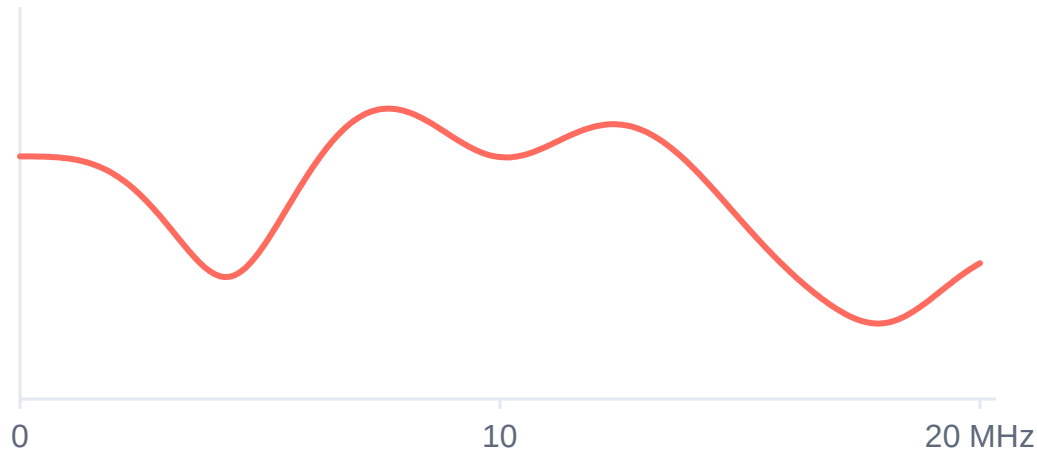
1 Why OFDM

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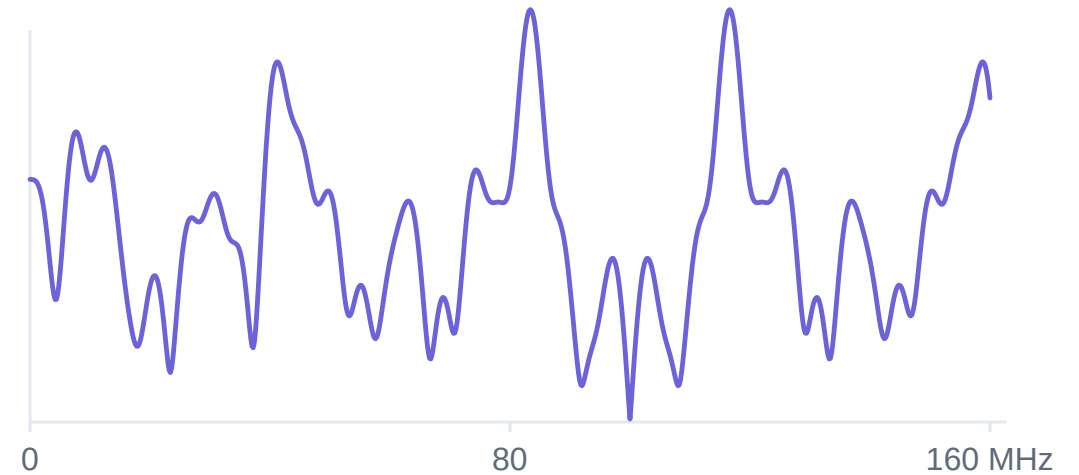
Six paths, and the gain is different at every frequency

SIX PATHS, OFFICE, 5 GHz — THE SAME CHANNEL AT TWO WIDTHS



20 MHz channel: two deep dips

12 dB between the best and worst frequency



160 MHz channel: fifteen dips

43 dB between the best and worst frequency

Two paths gave a regular ripple. Six give a curve with no pattern, and the receiver has to live on all of it.

Taps at 0, 25, 55, 95, 145 and 210 ns with gains 1.0, 0.68, -0.52, 0.40, 0.26, -0.16. This is a typical office profile, not a worst case.

A real channel is an irregular curve, not a number and not a ripple.

1 Why OFDM

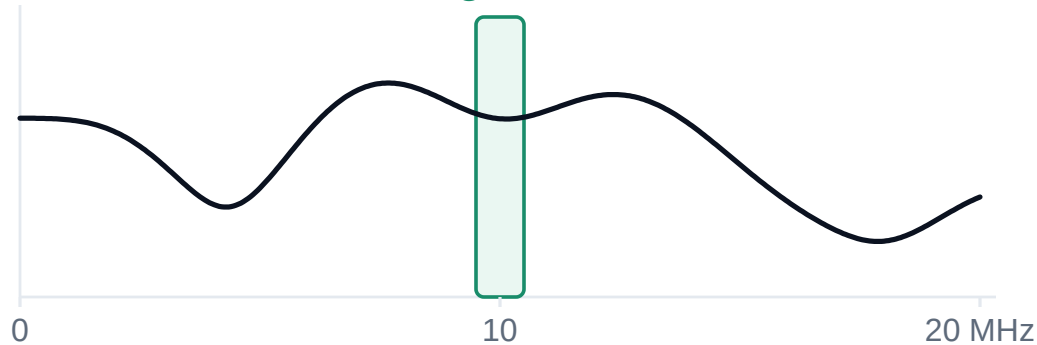
2 The channel

3 OFDM

Flat means one gain across the whole signal

THAT SAME SIX-PATH CHANNEL, LOOKED AT BY TWO DIFFERENT SIGNALS

a 1 MHz signal at 10 MHz



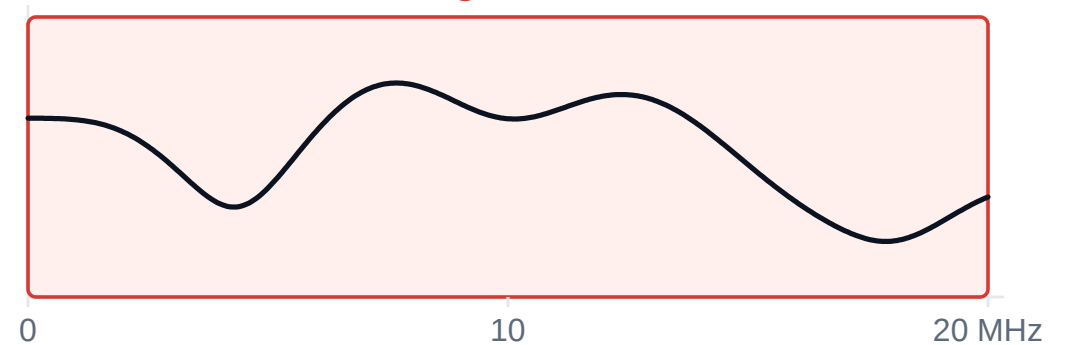
FLAT

the gain runs 1.65 to 1.70: 3% across it

Flat: one number is the whole channel.

Undoing it is one division.
That is the receiver Part 3 builds.

a 20 MHz signal, 0 to 20 MHz



NOT FLAT

the gain runs 0.52 to 1.99: four times

Not flat: no number is right.

Undoing it means undoing a filter,
the subtractions of Part 1.

The channel is fixed. The only thing a designer chooses is how wide the signal is.

Nothing about the channel changed. Flat is a statement about the signal, not about the channel.

Flat is what makes the channel a number instead of a filter.

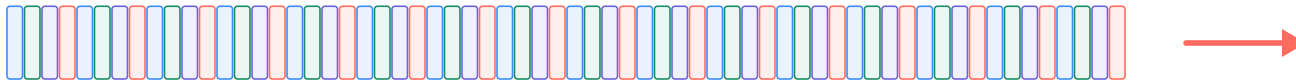
1 Why OFDM

2 The channel

3 OFDM

Cut the band, and every piece has to be flat

THE DECISION PART 1 ALREADY MADE



cut the wide signal into narrow pieces

The reason to cut it up is flatness: every piece has to see a single gain.

So the design question is a number:
how wide can one piece be before its gain stops being one number?

That width has a name.
It is the coherence bandwidth, B_c .

Everything in Part 3 — how many subcarriers, how wide each one is — is set by this one number.
Part 1 decided to split the band without saying how far. Flat is the test each piece has to pass, but a test is not a width.

Coherence bandwidth is the widest piece that is still flat.

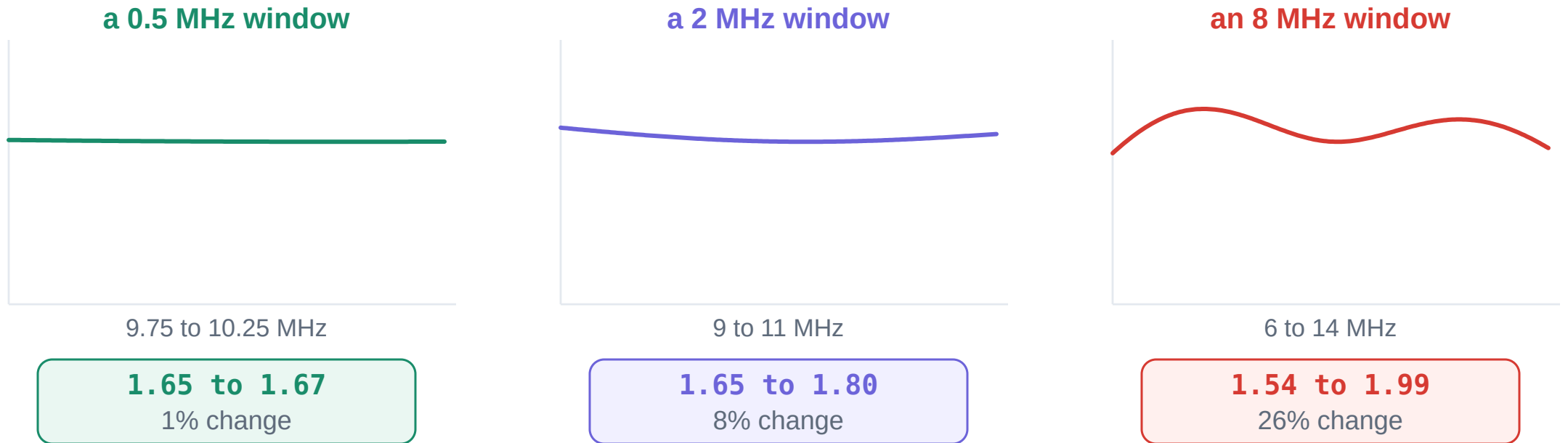
1 Why OFDM

2 The channel

3 OFDM

A narrow enough window sees one gain

MEASURING IT: THE SAME OFFICE CHANNEL, THROUGH THREE WINDOWS OF DIFFERENT WIDTH



The channel did not change. Only the width of the window did. Somewhere between 2 and 8 MHz it stops being one number.

All three plots are the office channel from the last slide, on the same vertical scale, centred on the same 10 MHz. The narrower the window, the less the gain moves inside it.

Coherence bandwidth is the widest window whose gain still counts as one number.

1 Why OFDM

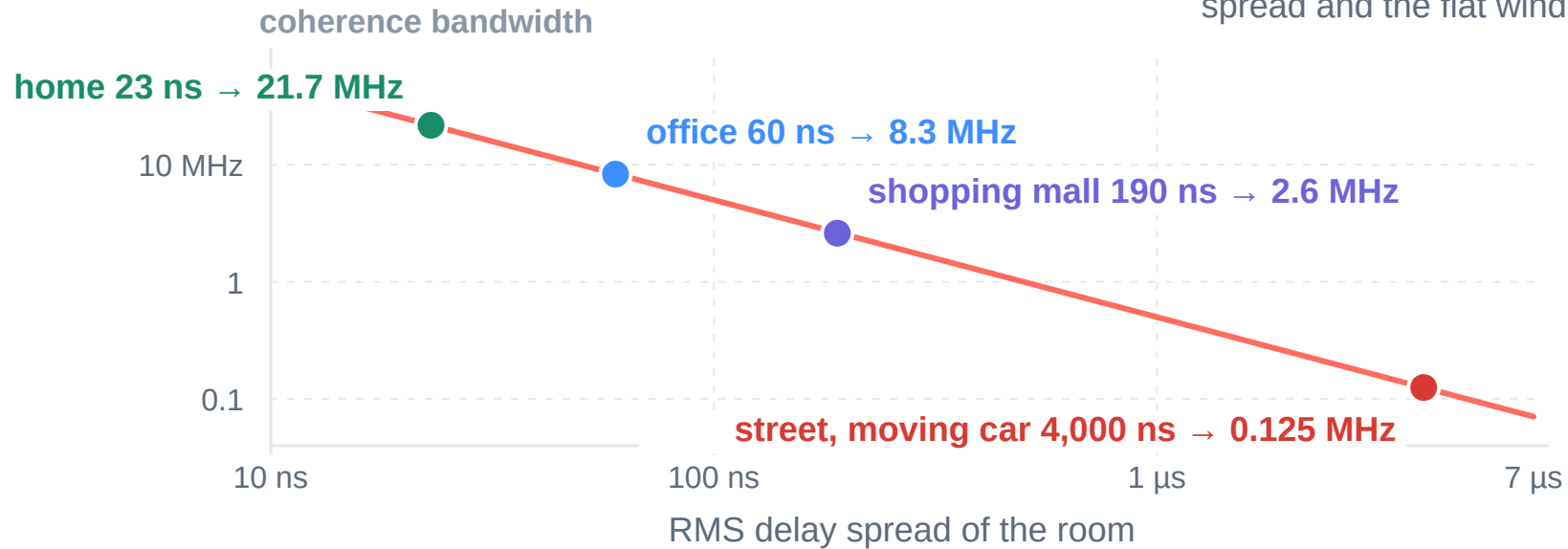
2 The channel

3 OFDM

Coherence bandwidth comes straight from the delay spread

$$B_c \approx 1 / (2 \tau_{\text{rms}})$$

A long delay spread means the gain changes quickly across frequency, so the flat window is narrow. Halve the delay spread and the flat window doubles.



IT IS A STATISTICAL STATEMENT

B_c is the width over which two gains are still similar, averaged over many rooms.

Textbooks put 1 to 50 in that denominator.

Both axes are logarithmic, so the relation is a straight line of slope -1 . The delay-spread figures are the measured medians from Part 1.

A few megahertz indoors, a fraction of one outdoors.

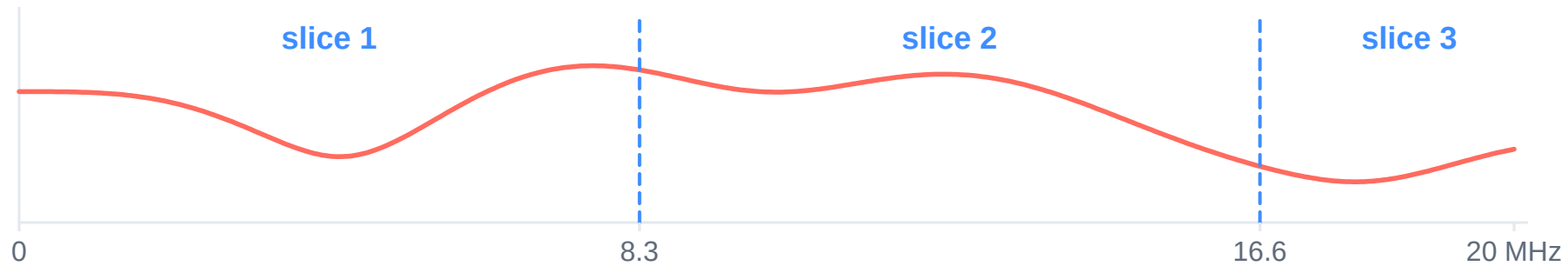
1 Why OFDM

2 The channel

3 OFDM

How many separate gains does one channel have?

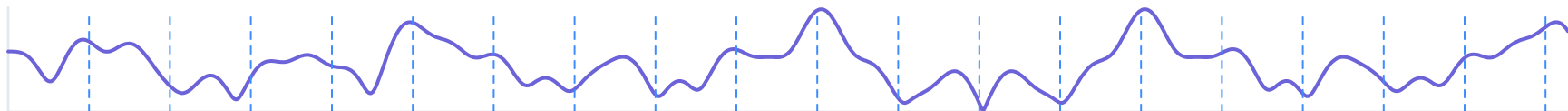
A 20 MHz CHANNEL IN AN OFFICE, WHERE B_c IS 8.3 MHz



Knowing the gain in slice 1 tells you almost nothing about slice 2. That is what B_c measures.

$$\text{independent gains} = \text{channel width} \div B_c$$

THE SAME OFFICE, BUT A 160 MHz CHANNEL



$160 \div 8.3 = \text{nineteen slices}$, so nineteen gains to measure and track

This is the number the receiver has to estimate and, in the MIMO lecture, feed back. It is not the number of subcarriers, and a slice is not a subcarrier — the next slide shows why.

A wider channel does not just carry more; it carries more things to measure.

1 Why OFDM

2 The channel

3 OFDM

● — Not flat yet

A piece one B_c wide still swings by half

THE SAME 20 MHz CHANNEL, CUT INTO PIECES FOUR DIFFERENT WAYS

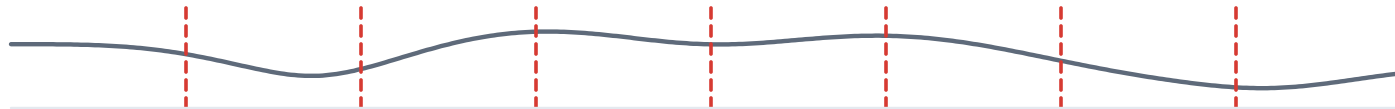
one B_c wide
8.3 MHz



MEDIAN SWING INSIDE ONE PIECE

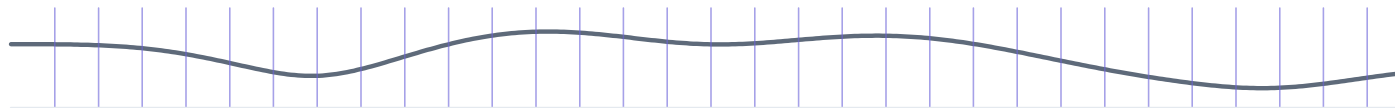
gain swings 58%

$B_c / 3$
2.5 MHz



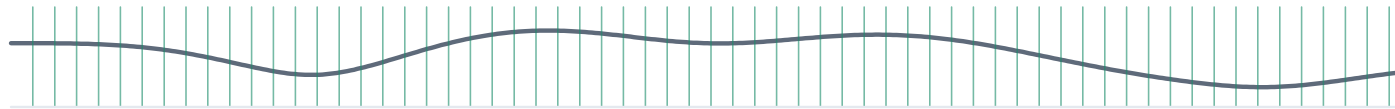
gain swings 37%

$B_c / 13$
625 kHz



gain swings 10%

$B_c / 27$
312.5 kHz



gain swings 5%

B_c is the scale on which the channel changes. It is not the width of a piece.

Cut at B_c and the gain inside one piece still runs from 0.7 to 1.9. You have to keep cutting. How far is a design choice, and Part 3 works it out from the constellation.

Divide the band by B_c to count gains, not to size a subcarrier.

1 Why OFDM

2 The channel

3 OFDM

Every channel a modern radio uses is frequency selective

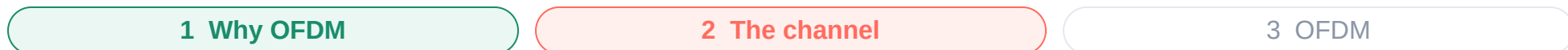
CHANNEL WIDTH $\div$ B_c , FOR FOUR WIDTHS AND FOUR PLACES

	20 MHz	40 MHz	80 MHz	160 MHz
home B~green 22 MHz	1 slice	2 slices	4 slices	7 slices
office B~blue 8.3 MHz	2 slices	5 slices	10 slices	19 slices
mall B~violet 2.6 MHz	8 slices	15 slices	31 slices	62 slices
street B _c 0.125 MHz	160 slices	320 slices	640 slices	1280 slices

Green is a channel a single-carrier receiver could handle. Everything else has to be split.

One cell is the count from the last slide, worked out for every combination. A 160 MHz channel in a shopping mall carries 62 separate gains.

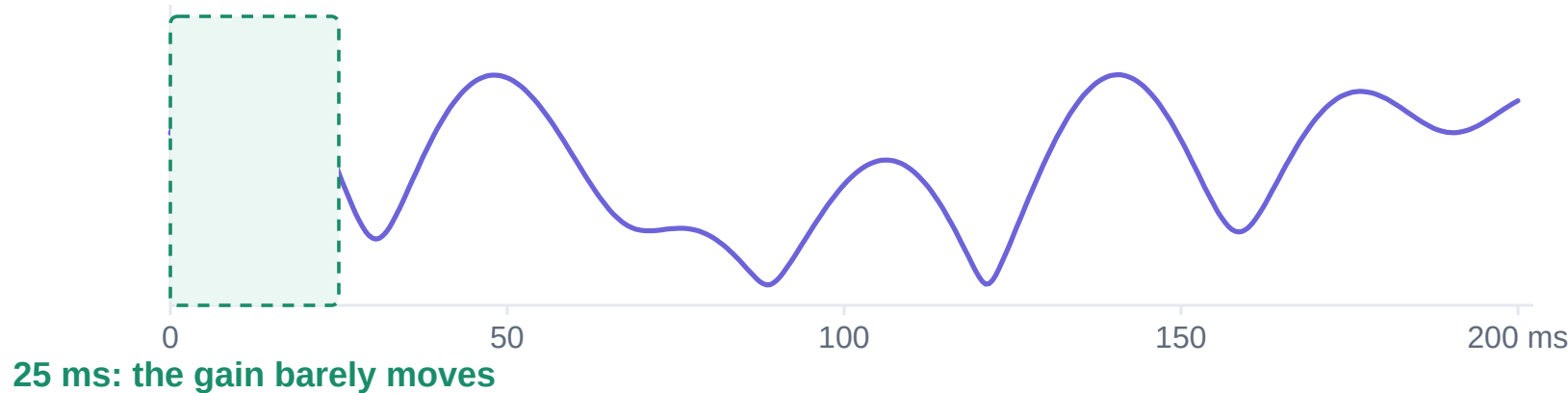
This is why every modern standard is an OFDM standard.



- The channel also moves

A measured channel goes stale as soon as anything moves

THE GAIN AT ONE FREQUENCY, WHILE THE RECEIVER WALKS AT 1 m/s



$$T_c \approx 0.42 \lambda / v$$

a wavelength is 6 cm at 5 GHz,
so 3 cm of movement turns
every path by half a cycle

a person walking, 1 m/s

channel holds still for 25 ms

a car, 30 m/s

channel holds still for 0.85 ms

A Wi-Fi packet lasts under 5 ms. Indoors the channel holds still for the whole packet. In a car it does not.

Doppler is v/λ , and coherence time is about 0.42 over that. This is why a packet re-sends training symbols, and why the MIMO lecture cares how old a channel report is.

Measure the channel, and the measurement has a shelf life.

1 Why OFDM

2 The channel

3 OFDM

Four things the channel forces on any design

The gain changes with frequency

so measure and correct it frequency by frequency

One symbol lands on the next

so leave a gap longer than the delay spread

Some frequencies are almost dead

so spread each codeword across many frequencies

The channel goes stale in milliseconds

so re-measure inside every packet

OFDM is one design that satisfies all four at once, and it does so with an FFT rather than a filter.

Part 3 builds it, one decision at a time.

1 Why OFDM

2 The channel

3 OFDM

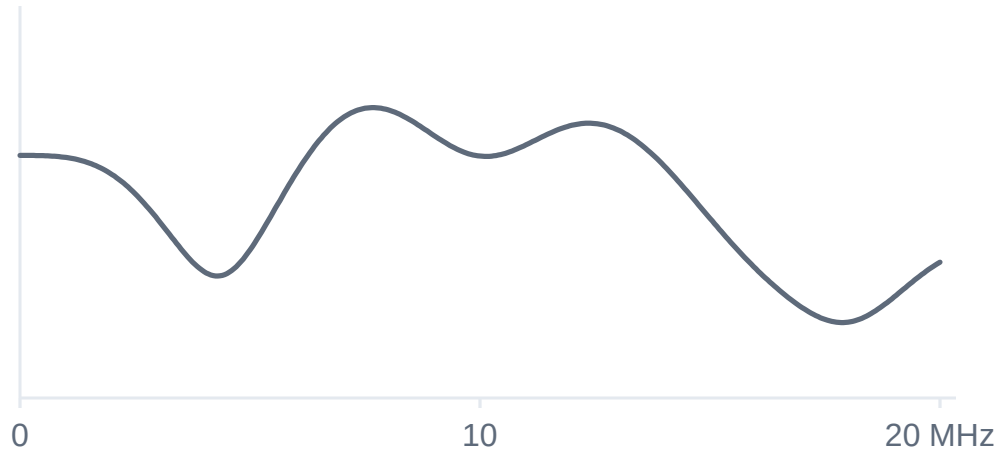
How does OFDM turn one channel into many?

03

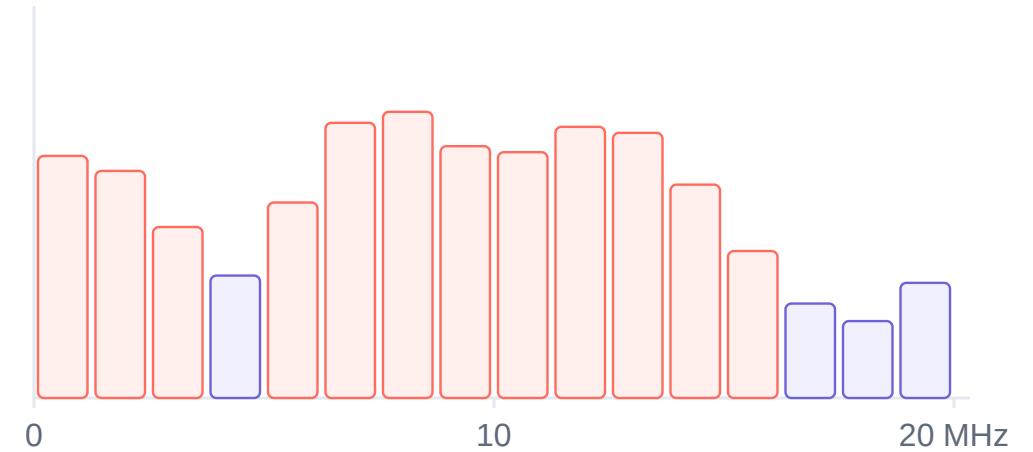
OFDM

Split the band into hundreds of pieces narrow enough that each one is flat, pack them so they overlap without interfering, generate them all with a single FFT, and put a guard in front of every symbol.

Cut the band into pieces narrow enough to be flat



ONE 20 MHz SIGNAL
one channel with many different gains



16 NARROW SIGNALS
each one has a single gain

Nothing about the channel changed. What changed is the width of the signal we ask each piece of the receiver to handle.

A narrow enough signal sees one number, not a curve.

1 Why OFDM

2 The channel

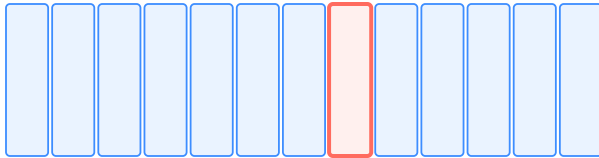
3 OFDM

• ——— What a subcarrier is

One sinusoid, one symbol long, carrying a few bits

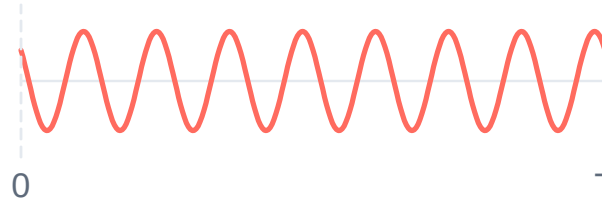
ONE SUBCARRIER, LOOKED AT THREE WAYS

IN FREQUENCY



one slot of the band, Δf wide
the 8th of 13 here, the 8th of 64 in 802.11a

IN TIME



a sinusoid at that frequency,
held for exactly one symbol T

WHAT SETS IT



its amplitude and phase are
the numbers being sent

A subcarrier is one sinusoid: its own frequency, one symbol long, carrying one QAM symbol.

All N are added together over that same T, and leave the antenna as one waveform.

Nothing is ever sent on a subcarrier by itself. It is a way of describing part of one signal.

Subcarrier, tone and FFT bin are three names for the same thing, indexed by k.

N subcarriers, N QAM symbols, one transmitted waveform.

1 Why OFDM

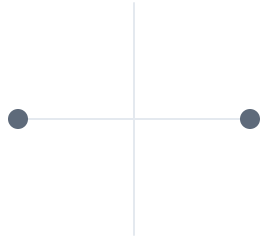
2 The channel

3 OFDM

The constellation decides how many bits each one carries

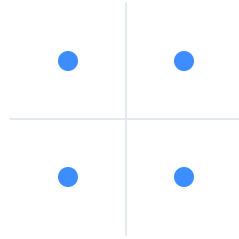
ONE SUBCARRIER, ONE OFDM SYMBOL: HOW MANY BITS?

BPSK



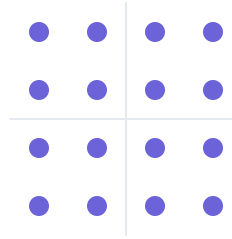
1 bit
per symbol

QPSK



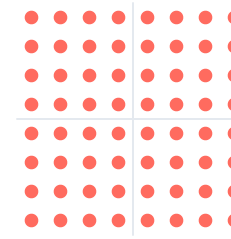
2 bits
per symbol

16-QAM



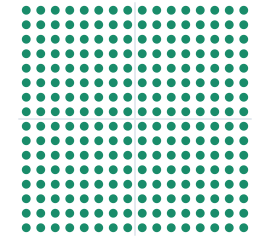
4 bits
per symbol

64-QAM



6 bits
per symbol

256-QAM



8 bits
per symbol

bits per OFDM symbol = $N \times \text{bits per subcarrier}$

802.11a: 48 data subcarriers $\times$ 6 bits = 288 bits every 4 μ s.

Every subcarrier can use a different constellation, and real systems do exactly that.

More points in the picture, more bits, and less room for error.

1 Why OFDM

2 The channel

3 OFDM

Bits go in one at a time and leave all at once

SERIAL IN: ONE BIT AT A TIME



ONE BLOCK = 16 BITS

8 subcarriers × 2 bits each

10 00

• •

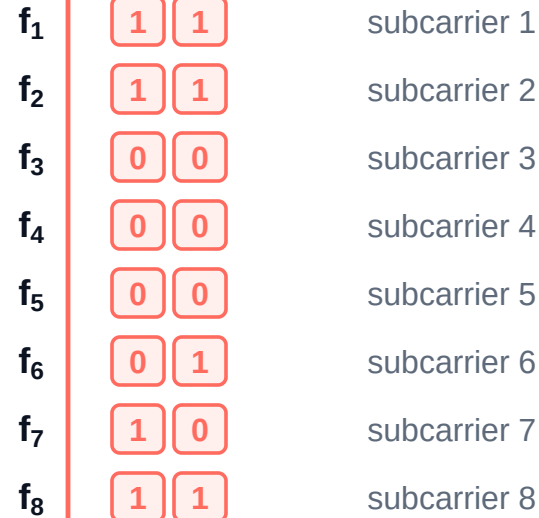
• •

11 01

QPSK

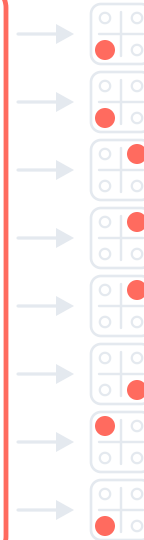
2 bits, one point

ONE BLOCK: 8 SUBCARRIERS × 2 BITS



block 2 · all 16 bits placed

PARALLEL OUT: ONE POINT PER SUBCARRIER



ALL 8 LEAVE TOGETHER: ONE OFDM SYMBOL

With 64 subcarriers and QPSK, 128 bits become 64 points, and all 64 leave together.

Drawn with 8 subcarriers and QPSK so the rows are visible.

One bit at a time in. A whole OFDM symbol out, in one instant.

1 Why OFDM

2 The channel

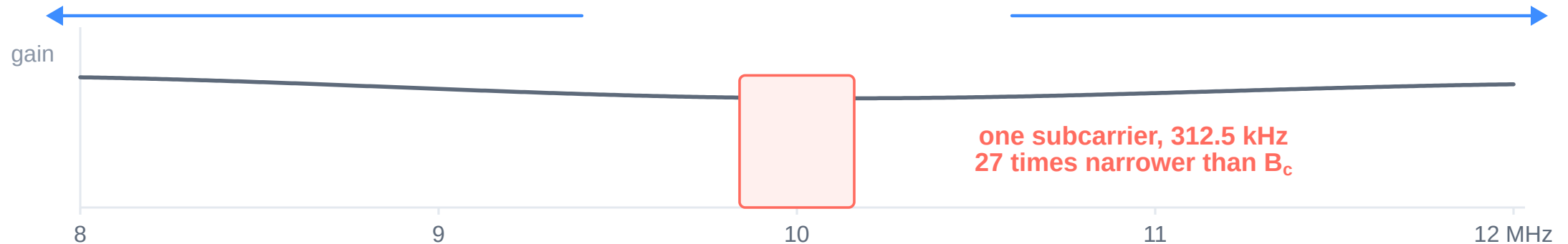
3 OFDM

- How narrow is narrow enough

A subcarrier must be far narrower than B_c

subcarrier spacing $\Delta f = \text{bandwidth} / N$

$B_c = 8.3 \text{ MHz}$, wider than this whole plot



Across that 312.5 kHz the gain moves by 0.6% here, and by 2.9% at the median point of this channel.

A subcarrier the width of B_c is not flat.

B_c is where the gain has already changed a lot — that is how it is defined.

802.11a chose 27 times narrower than B_c . 802.11ax chose 106 times narrower, which is what 1024-QAM needs.

The receiver divides by one number per subcarrier, so any gain change inside a subcarrier is an error no division can remove.

$\Delta f = B_c$ is the point where flatness fails, not the target.

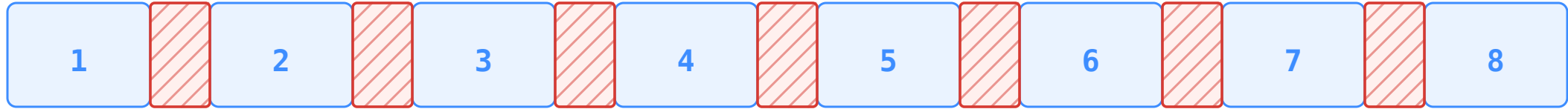
1 Why OFDM

2 The channel

3 OFDM

Separate channels with guard bands and you lose a quarter

THE OBVIOUS WAY: PUT A GUARD BAND BETWEEN NEIGHBOURS



guard band
nothing is sent here

8 subcarriers and 7 guard bands:
27% of the span carries nothing

More subcarriers means more gaps, so the waste does not shrink.

The gap exists only because a real filter does not stop at its edge.

**Is there a way to shrink the guard band —
or to remove it completely?**

A real filter does not stop at its edge, so it needs the gap.

Guard bands are the cost of separating signals with filters.

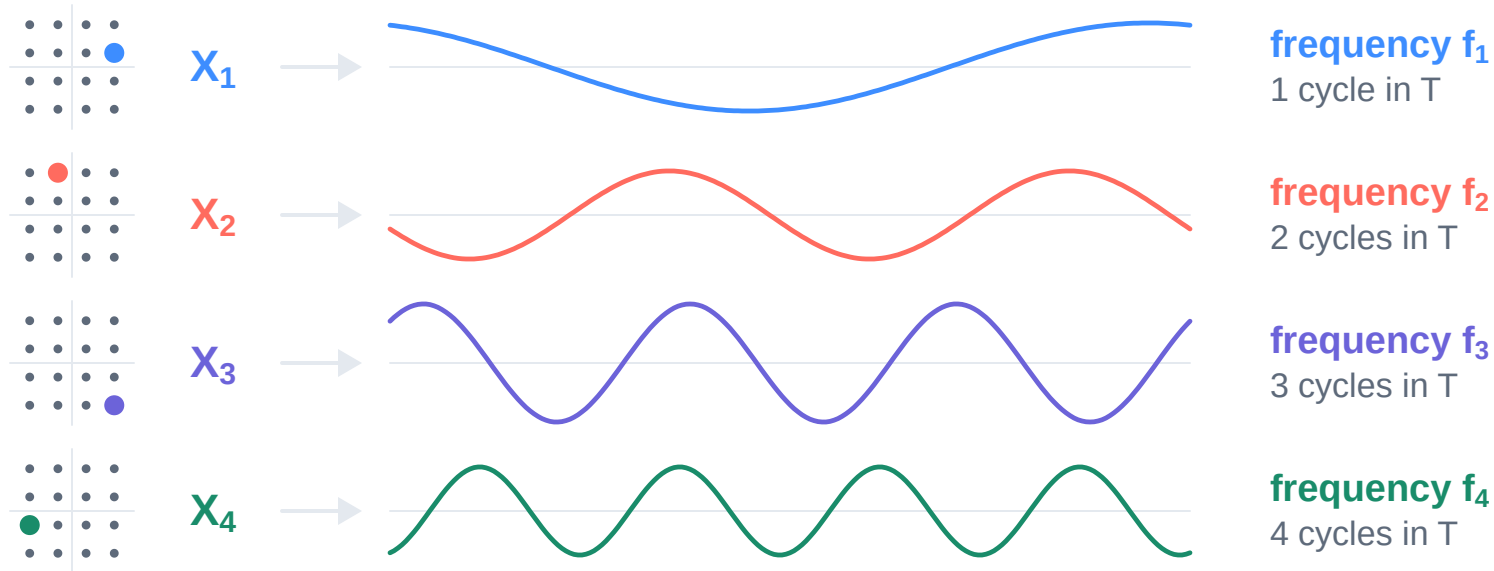
1 Why OFDM

2 The channel

3 OFDM

Each subcarrier carries one complex number, X_k

FOUR SUBCARRIERS, AND THE FOUR NUMBERS THEY CARRY



X_k

is one QAM symbol:
**an amplitude and
a phase.**

It sets the size and
the shift of sinusoid k .

Four QAM symbols, four sinusoids. Everything below is about getting them back.

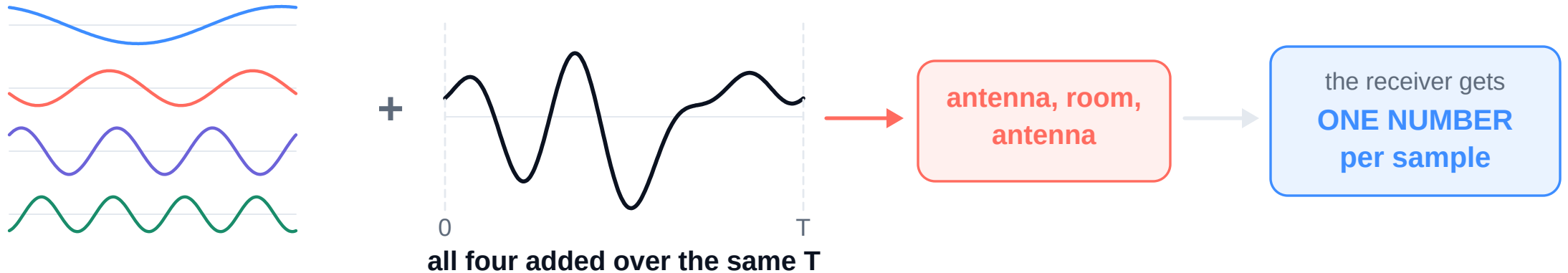
1 Why OFDM

2 The channel

3 OFDM

The four sinusoids leave as one waveform

THE TRANSMITTER ADDS THEM. ONLY THE SUM LEAVES THE ANTENNA.



We receive the sum. We want back what each of f_1, f_2, f_3, f_4 was carrying.

Whether that is possible depends on how we pick them.

The sum is the signal. Getting the parts back is the whole problem.

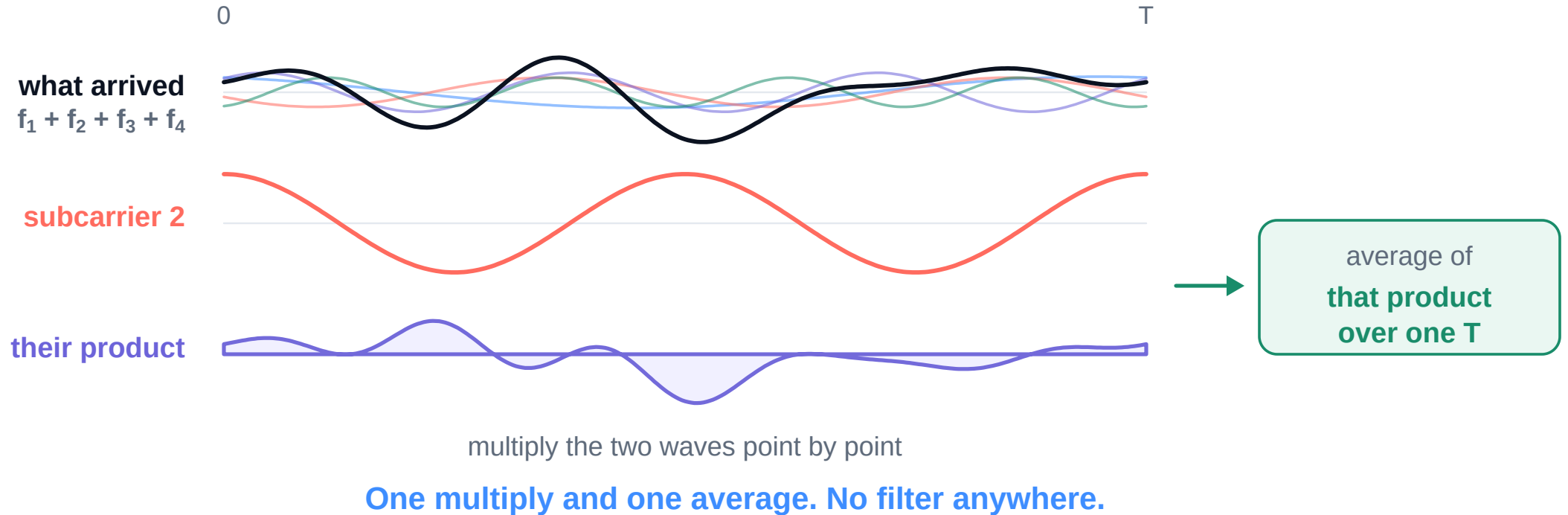
1 Why OFDM

2 The channel

3 OFDM

Multiply by the one you want, then average

TO GET X_2 : MULTIPLY THE ARRIVING WAVE BY SUBCARRIER 2, THEN AVERAGE



This is a correlator, and it is what one FFT bin computes.

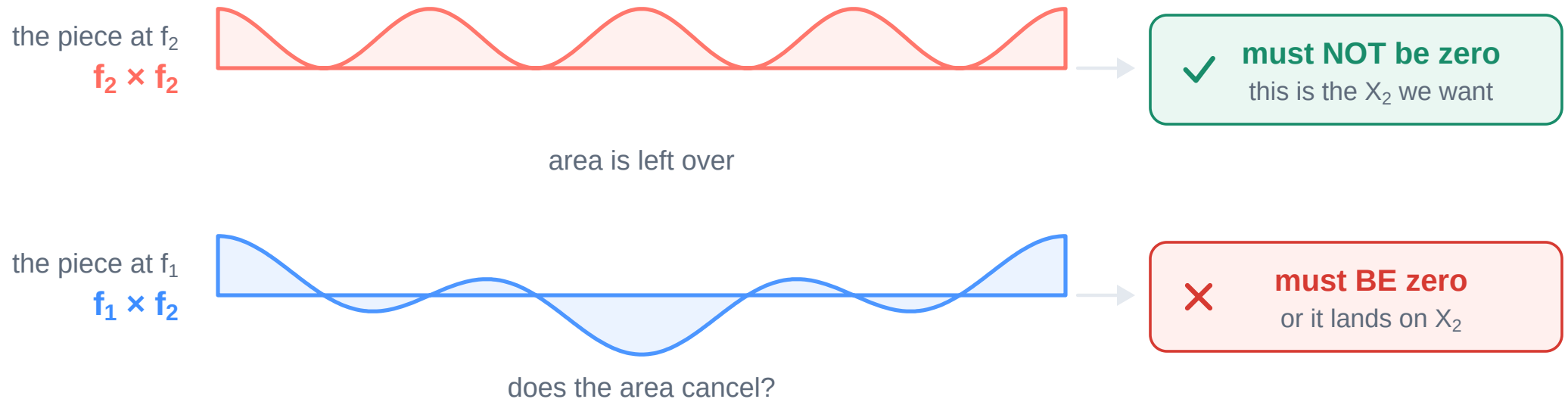
1 Why OFDM

2 The channel

3 OFDM

Keep one product, kill all the others

MULTIPLY BY f_2 . WHAT DOES EACH PIECE TURN INTO?



The top one is what we keep. The bottom one has to be zero, or it lands on top of it.

Both pictures are one piece of the arriving wave multiplied by f_2 , over one symbol.

So: when is the area under a product zero, and when is it not?

1 Why OFDM

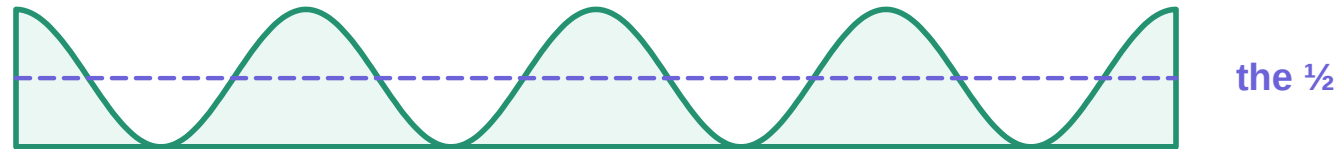
2 The channel

3 OFDM

A wave times itself has a constant in it

THE PIECE AT f_2 , MULTIPLIED BY f_2

$$\cos(2\pi ft) \times \cos(2\pi ft) = \frac{1}{2} + \frac{1}{2} \cos(2\pi \cdot 2f \cdot t)$$



the whole product sits above the axis

The $\frac{1}{2}$ is a constant. A constant does not average away.
So this product never averages to zero, whatever f is.

The first requirement is free. Nothing has to be arranged for it.

The wanted subcarrier always survives. Now kill the others.

1 Why OFDM

2 The channel

3 OFDM

A product of two different subcarriers is two waves

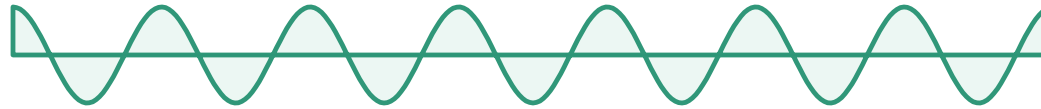
THE PIECE AT f_5 , MULTIPLIED BY f_2

$$\cos(2\pi f_5 t) \times \cos(2\pi f_2 t) = \frac{1}{2} \cos(2\pi(f_5 - f_2)t) + \frac{1}{2} \cos(2\pi(f_5 + f_2)t)$$

the difference frequency



the sum frequency



**No constant this time. Just two waves.
So this averages to zero only if both waves do.**

Two cosines multiplied give one at the difference frequency and one at the sum.

The second requirement is not free. It has to be arranged.

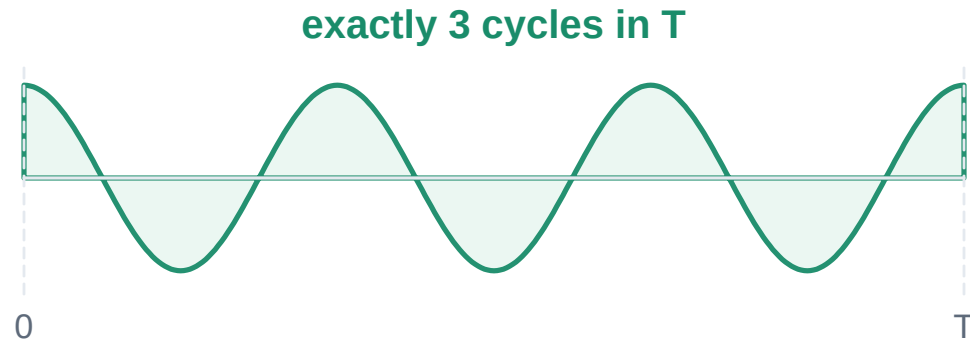
1 Why OFDM

2 The channel

3 OFDM

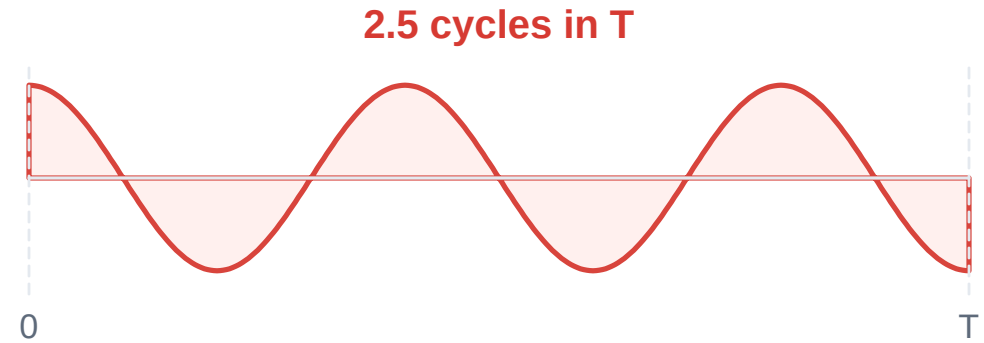
A wave averages to zero over whole cycles only

SO: WHEN DOES A WAVE AVERAGE TO ZERO OVER THE WINDOW T ?



the areas cancel

average = ZERO



half a cycle is left over

average = NOT ZERO

A wave averages to zero over T if, and only if,
a whole number of its cycles fits in T .

Area above the axis counts positive, below counts negative.

Whole number of cycles in T , or the average is not zero.

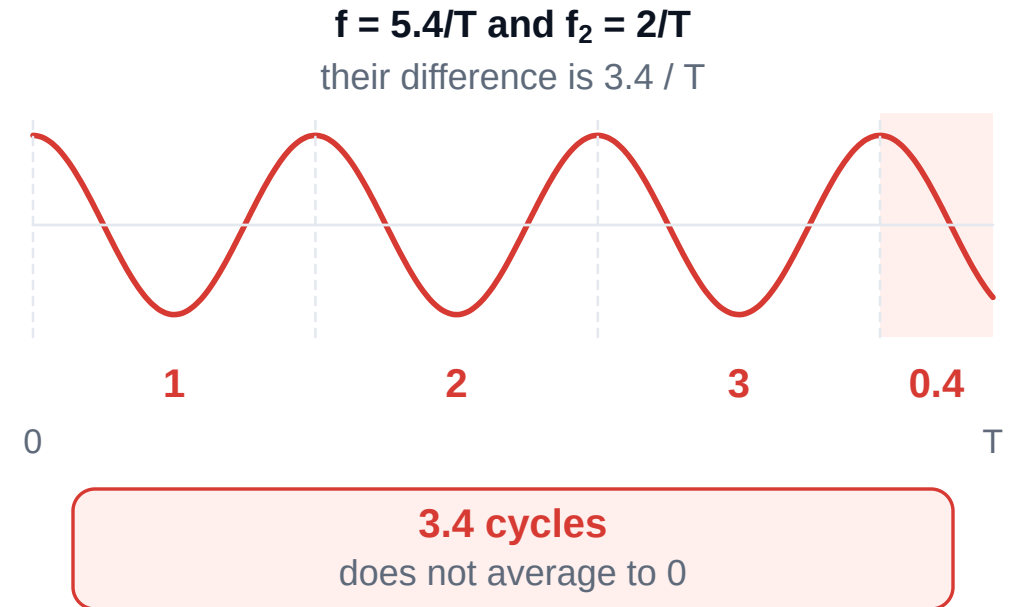
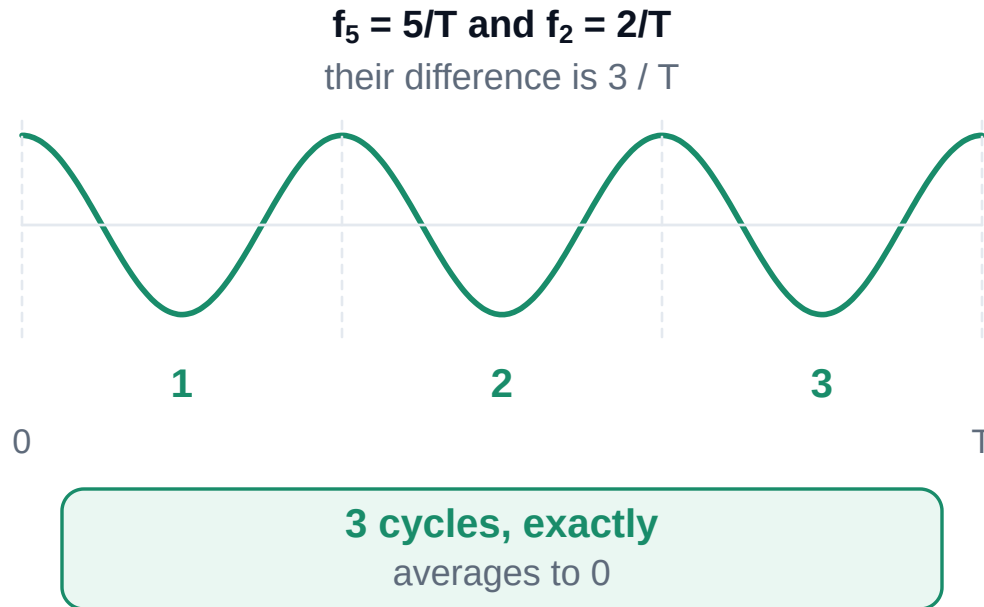
1 Why OFDM

2 The channel

3 OFDM

The difference wave must finish inside the window

THE DIFFERENCE WAVE, DRAWN OVER THE WINDOW T , WITH ITS CYCLES COUNTED



The wave at $f_k - f_m$ has to finish a whole number of cycles inside T . Nothing may be left over.

The numbers under the axis count the cycles of the difference wave, not of either subcarrier.

Whole cycles, or a leftover piece that nothing cancels.

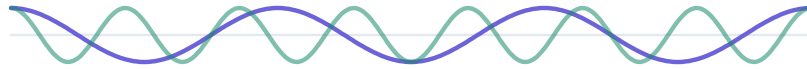
1 Why OFDM

2 The channel

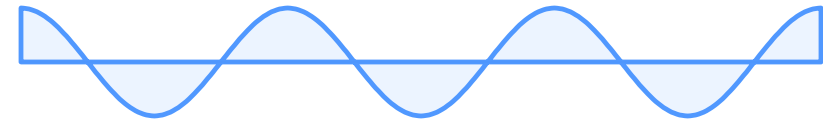
3 OFDM

What the product has to satisfy to vanish

WHAT THE LAST TWO SLIDES SAID



$f_k \times f_m$ is a difference wave and a sum wave



a wave averages to zero only over whole cycles



the wave at $f_k - f_m$ must finish a whole number of cycles inside T

$(f_k - f_m) \times T$ is a whole number

the wave at $f_k + f_m$ must too

$(f_k + f_m) \times T$ is a whole number

“A whole number of cycles in T ” is what both lines mean.

Now find the frequencies that satisfy both, for every pair.

1 Why OFDM

2 The channel

3 OFDM

• The condition

Every frequency is a whole multiple of $1 / T$

ADD THE TWO CONDITIONS TOGETHER

$$(f_k - f_m) \times T \quad \text{whole number of cycles in } T$$

$$(f_k + f_m) \times T \quad \text{whole number of cycles in } T$$

$$2 f_k \times T \quad \text{whole number of cycles, for every } k$$

$f_k \times T$ is how many cycles subcarrier k fits in T — it has to be a whole one

$$f_k = k / T \text{ for } k = 1, 2, 3, \dots$$

the simplest set that works, and the tightest

Subcarrier k fits exactly k cycles into one symbol. That is the whole rule.

Adding the two conditions removes f_m , so each frequency is pinned on its own, not just relative to its neighbour.

Neighbours are $1/T$ apart, so $\Delta f = 1 / T$.

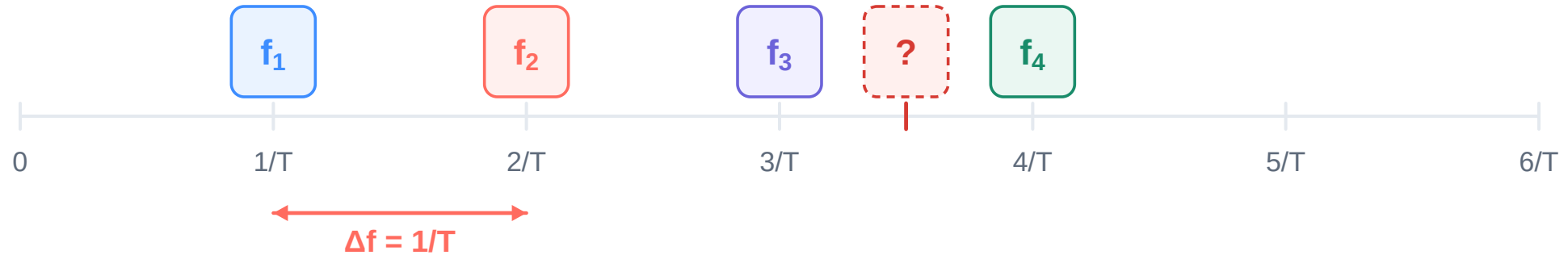
1 Why OFDM

2 The channel

3 OFDM

The subcarriers sit on ticks $1/T$ apart, and nowhere else

THE ALLOWED FREQUENCIES, DRAWN ON THE FREQUENCY AXIS



a frequency between the ticks
is 2.5 cycles away from f_1
and leaks into every other one

The frequencies form a grid, and the grid step is $1/T$.

Pack them onto consecutive ticks and nothing is wasted.

Orthogonality is a grid in frequency, tied to the symbol length.

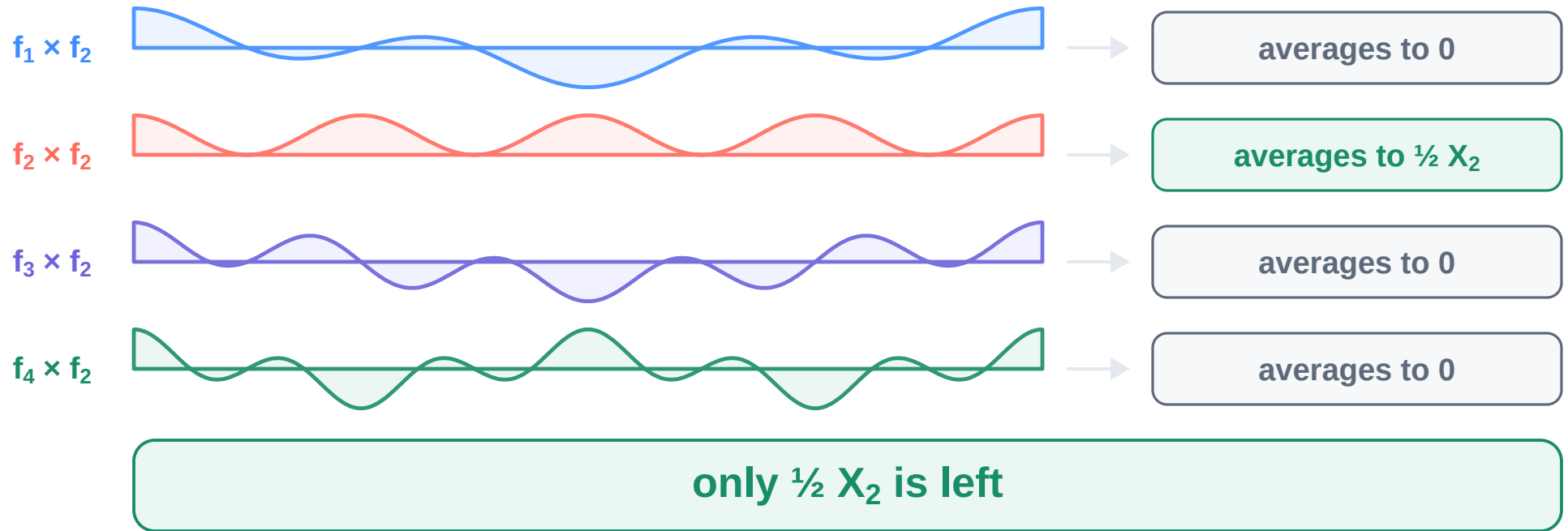
1 Why OFDM

2 The channel

3 OFDM

Three products average away, one does not

THE ARRIVING WAVE IS FOUR PIECES. MULTIPLY EACH BY SUBCARRIER 2.



Only the second row has area left over, because only it has a constant in it.

One multiply, one average, and X_2 comes back on its own.

1 Why OFDM

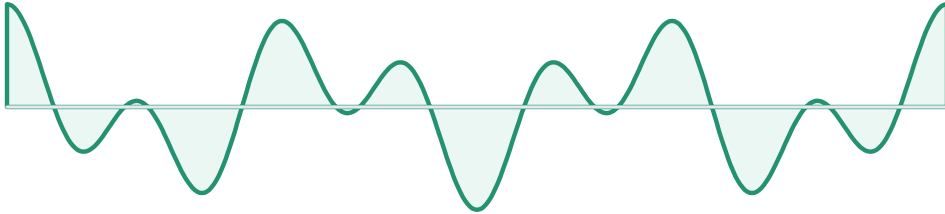
2 The channel

3 OFDM

Spacing and symbol length are the same choice

THE SAME PRODUCT WHEN THE SPACING IS NOT EXACTLY $1/T$

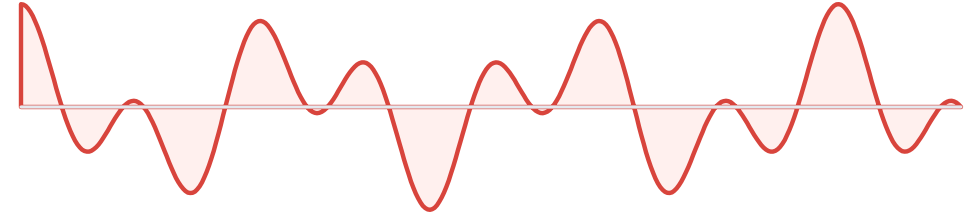
spacing = $1/T$



the two halves have 3.00 and 7.00 cycles

average = ZERO

spacing = $1.15/T$



the two halves have 3.45 and 8.05 cycles

average = NOT ZERO

subcarrier spacing $\Delta f = 1/T$ exactly

Pick T from the delay spread, and the spacing is decided for you. There is nothing left to tune.

The leftover lands on top of the wanted subcarrier, inside the number the FFT returns, where no equaliser can reach it.

$\Delta f = 1/T$ is what makes an OFDM system OFDM.

1 Why OFDM

2 The channel

3 OFDM

Spacing comes from T; bandwidth comes from N

THE CONDITION FIXES THE SPACING. THE BANDWIDTH IS A SEPARATE CHOICE.

$$\Delta f = 1 / T$$

spacing between neighbours,
set by the symbol length

$$B = N \times \Delta f$$

total bandwidth, set by how
many you stack side by side

Two different things. Pick any two of B, N, Δf , T and the other two follow.

STANDARD	B	N	Δf	$T = 1/\Delta f$
802.11a	20 MHz	64	312.5 kHz	3.2 μ s
802.11ax, 160 MHz	160 MHz	2,048	78.125 kHz	12.8 μ s
LTE	20 MHz	2,048	15 kHz	66.7 μ s

The condition is about one subcarrier and T. The bandwidth is just how many of them there are.

$\Delta f = 1/T$ is orthogonality. $B = N \cdot \Delta f$ is bookkeeping.

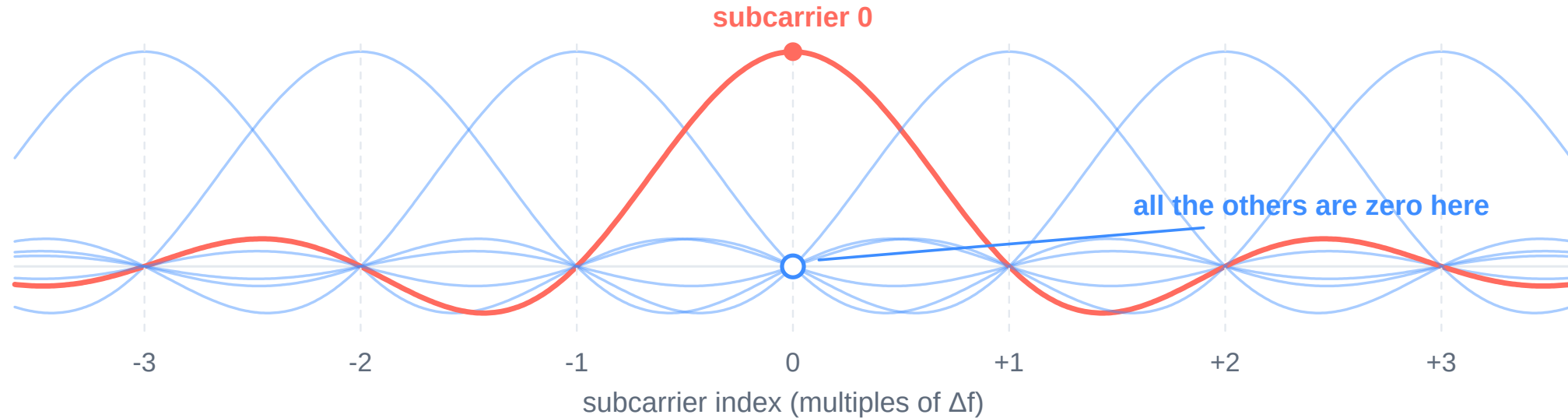
1 Why OFDM

2 The channel

3 OFDM

Overlapping spectra that are still separable

THE SAME FACT SEEN IN FREQUENCY: EACH SUBCARRIER IS A SINC



The spectra overlap heavily, yet at every subcarrier centre only one is non-zero.

A rectangle of length T in time is a sinc in frequency, and its zeros are spaced $1/T$ apart — exactly Δf .

Sampling at the centres picks out one subcarrier and nothing else.

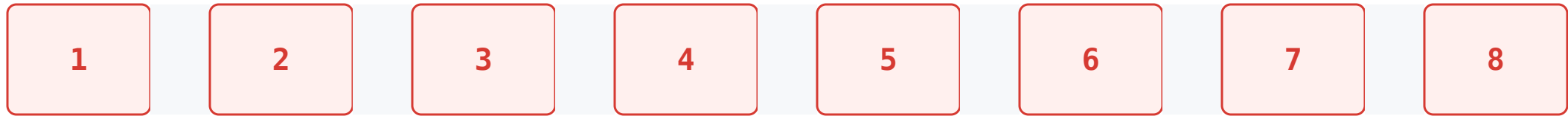
1 Why OFDM

2 The channel

3 OFDM

Removing the guard bands returns the spectrum they cost

WITH GUARD BANDS



8 subcarriers in this span

WITH ORTHOGONAL OVERLAP



11 subcarriers in the same span

Both rows span the same spectrum. Removing the gaps fits three more subcarriers into it.

OFDM packs subcarriers as tightly as separability allows.

1 Why OFDM

2 The channel

3 OFDM

Overlap, and separate them by arithmetic instead

THE QUESTION FROM FOUR SLIDES AGO

Is there a way to shrink the guard band — or to remove it completely?

1 The guard band is gone.

Neighbours overlap. The spacing Δf is the whole subcarrier, and nothing between them is wasted.

2 They are still separable.

At the centre of each subcarrier every other one is exactly zero. The receiver multiplies and averages. It never filters.

3 One constraint replaces the guard band.

$\Delta f = 1/T$ exactly, so the two radios must agree on frequency and on where the symbol starts.

Spectrum was the expensive thing. Clock accuracy is cheap. That is the whole trade.

Orthogonality replaced a spectrum cost with an accuracy requirement.

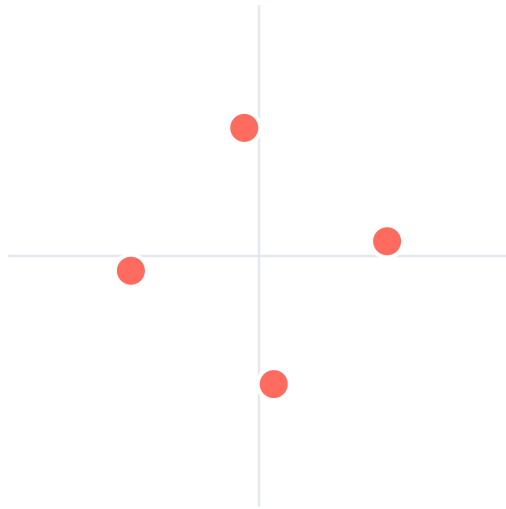
1 Why OFDM

2 The channel

3 OFDM

The receiver has Y_k , and cannot read it yet

AFTER THE FFT, ONE SUBCARRIER: THE FOUR QPSK POINTS AS THEY ARRIVE



Y_k , what the FFT gives back

The equaliser needs H_k . Nothing in Y_k says what it is, so the transmitter has to help.

Every subcarrier has its own H_k , so this has to be done 64 times, on every packet.

Which of the four
was actually sent?

From this picture alone: no idea.

The four points are all
shrunk and turned by
the same H_k .

SO, BEFORE ANY BITS

measure H_k

on every subcarrier,
then divide it out.

That measurement is
channel estimation.

Channel estimation is measuring H_k before decoding anything.

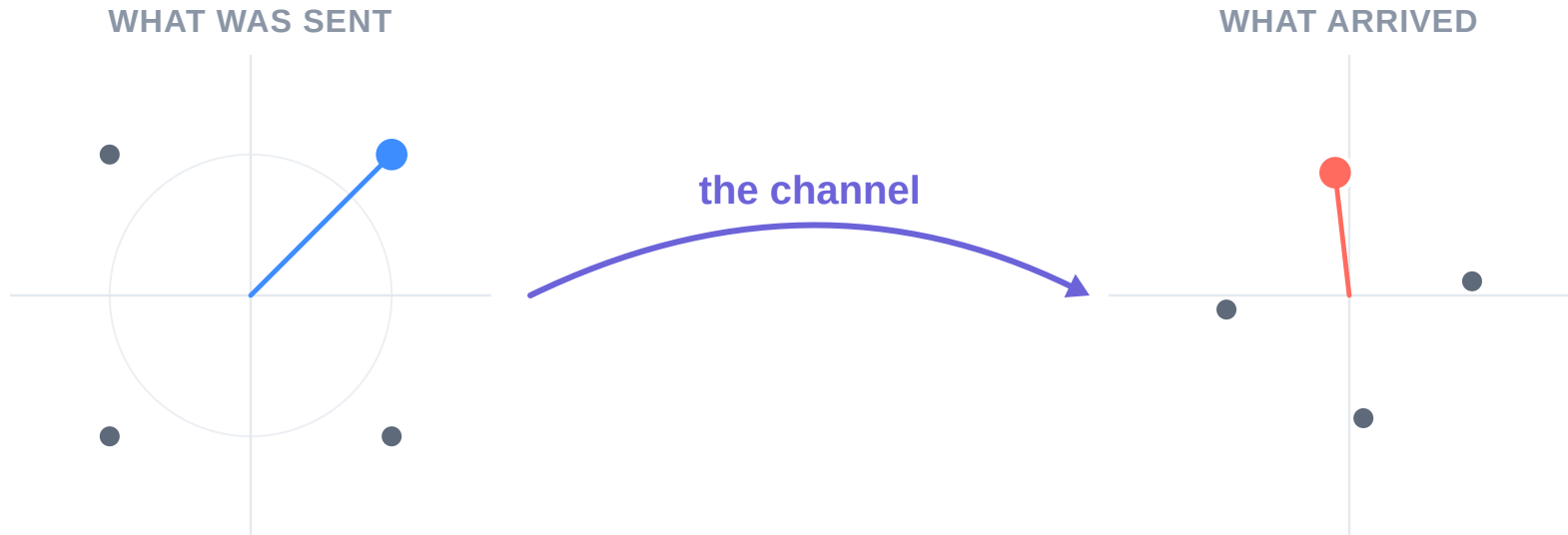
1 Why OFDM

2 The channel

3 OFDM

It moves the constellation point, and nothing else

ONE SUBCARRIER, ONE QPSK SYMBOL, BEFORE AND AFTER THE CHANNEL



The channel shrank the point and turned it.

That shrink and that turn are the whole channel, on this subcarrier.

On a flat subcarrier the channel cannot distort the shape. It can only scale and rotate.

One subcarrier, one gain, one phase.

1 Why OFDM

2 The channel

3 OFDM

Divide what arrived by what you sent

IF YOU KNOW WHAT YOU SENT, YOU CAN WORK OUT WHAT THE CHANNEL DID

$$H_k = Y_k \text{ received} / X_k \text{ sent}$$

$$H_k = |H_k| \times e^{j\angle H_k}$$

$|H_k|$

how much the point shrank or grew

0.62 here

$\angle H_k$

how far the point turned

52° here

One division per subcarrier, and the channel on it is known.

This is why a packet begins with symbols the receiver already knows: X_k has to be known for the division to mean anything.

H_k is one complex number: a size and an angle.

1 Why OFDM

2 The channel

3 OFDM

Sixty-four numbers, and they are the CSI

DO THAT ON ALL 64 SUBCARRIERS AND YOU HAVE 64 COMPLEX NUMBERS



These two curves together are the channel state information.

One complex number per subcarrier: a vector of 64.

CSI is the channel written down, one number per subcarrier.

1 Why OFDM

2 The channel

3 OFDM

Divide by H_k and the constellation snaps back

KNOWING H_k , UNDO IT: DIVIDE, ONE SUBCARRIER AT A TIME

$$\hat{X}_k = Y_k / H_k$$



The points land back where they started, and the bits can be read off.

48 divisions per OFDM symbol in 802.11a. A time-domain equaliser for the same channel needs a 10-tap filter on every sample.

OFDM replaced a filter design problem with N divisions.

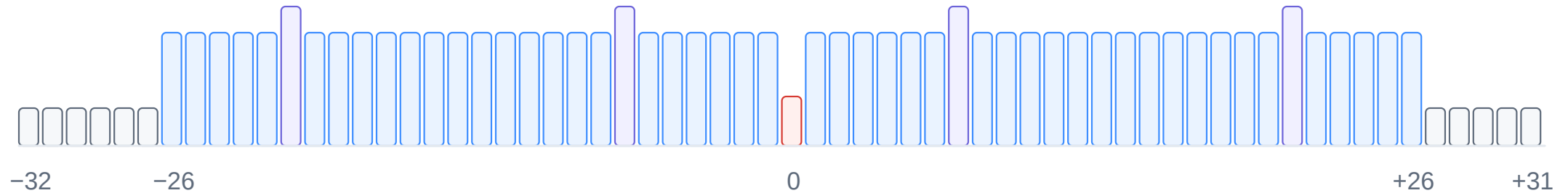
1 Why OFDM

2 The channel

3 OFDM

Four subcarriers carry no data and track the drift

THE 64 BINS OF AN 802.11a SYMBOL



48 data subcarriers

4 pilots at ± 7 and ± 21

1 null at DC

11 guard bins at the edges

The pilots carry a known sequence for the whole packet. Comparing what arrives on them with what was sent tracks the phase drift that builds up after the preamble, on every symbol.

The edges are left empty so the transmitted spectrum falls off before the neighbouring channel, and DC is left empty because receivers leak there.

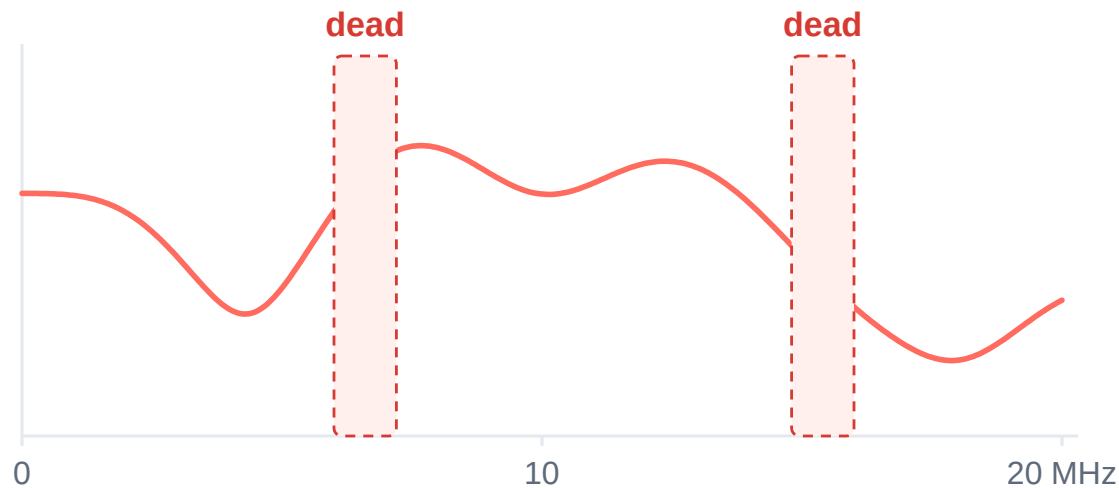
52 of 64 bins are used; 48 of those carry data.

1 Why OFDM

2 The channel

3 OFDM

Some subcarriers are useless, so no codeword may sit in one



dividing by a tiny H_k multiplies the noise

THE FIX: SPREAD EVERY CODEWORD OUT

without interleaving



one codeword in dark; the rest in grey

with interleaving



a notch now damages one bit of many codewords,
and the code repairs each of them

The interleaver in 802.11a sends adjacent coded bits to subcarriers far apart. Coding across frequency is what turns notches into recoverable loss.

OFDM needs coding across subcarriers to survive its own notches.

1 Why OFDM

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3 OFDM

The same five parameters in four different standards

PARAMETER	802.11a	802.11ax	LTE	5G NR $\mu=1$
channel width	20 MHz	160 MHz	20 MHz	100 MHz
FFT size	64	2,048	2,048	4,096
subcarrier spacing	312.5 kHz	78.125 kHz	15 kHz	30 kHz
symbol, without prefix	3.2 μ s	12.8 μ s	66.7 μ s	33.3 μ s
cyclic prefix	0.8 μ s	0.8 – 3.2 μ s	4.7 μ s	2.3 μ s
data subcarriers	48	1,960	1,200	3,276
prefix overhead	20%	5.9%	6.6%	6.6%

Four standards, one design. Only the numbers change, and they change with the cell size and the channel width.

Bigger cells push the prefix up; wider channels push the spacing down.

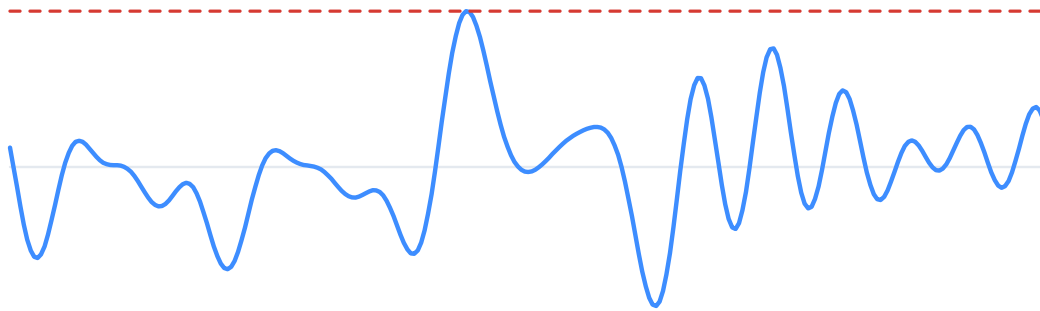
1 Why OFDM

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The sum of many subcarriers has a very high peak

ADDING N SUBCARRIERS SOMETIMES ADDS N PEAKS



peak is far above the average

N = 64, worst case

18.1 dB

N = 64, exceeded 0.01% of the time

11.3 dB

N = 2,048, worst case

33.1 dB

N = 2,048, exceeded 0.01% of the time

12.3 dB

The amplifier must stay linear up to the peak,
so it runs well below its rated power.

That costs battery life and range.

Worst case is $10 \log_{10} N$, when every subcarrier lines up. It almost never happens, but the amplifier still has to be designed for what does happen.

OFDM trades amplifier efficiency for a simple receiver.

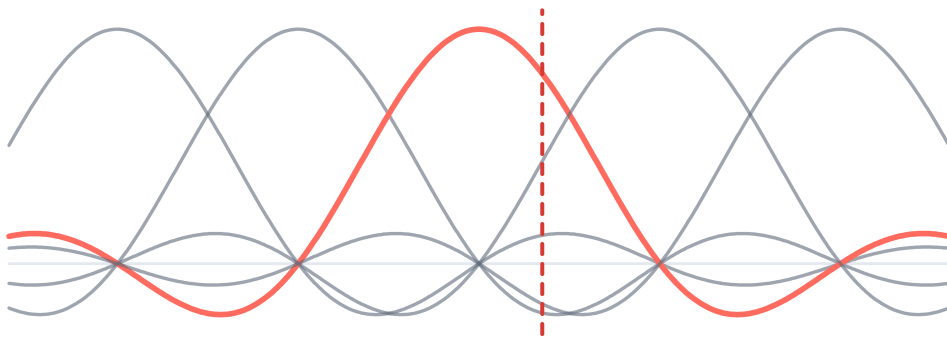
1 Why OFDM

2 The channel

3 OFDM

A frequency error destroys the orthogonality

IF THE TWO RADIOS DISAGREE ON THE CARRIER FREQUENCY
the receiver samples here



the wanted subcarrier is smaller, and the
others are no longer zero

oscillator tolerance in 802.11

± 20 ppm

that is, at 5 GHz

± 100 kHz

as a share of the 802.11a subcarrier

0.32

as a share of the 802.11ax subcarrier

1.28

Narrower subcarriers are more fragile. The preamble must estimate
and remove the offset before the FFT runs.

Sampling off centre lets every other subcarrier leak in. This is intercarrier interference, and it grows with the square of the offset.

The finer the subcarrier, the tighter the frequency lock must be.

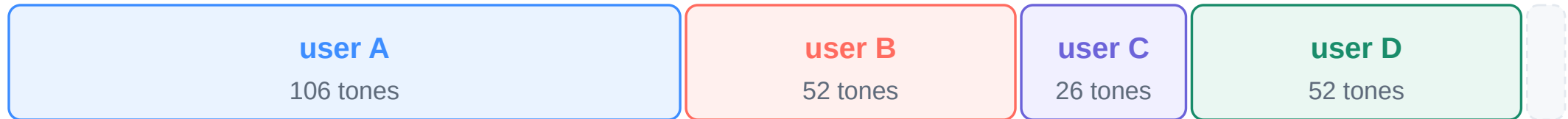
1 Why OFDM

2 The channel

3 OFDM

Once the band is in pieces, pieces can go to different users

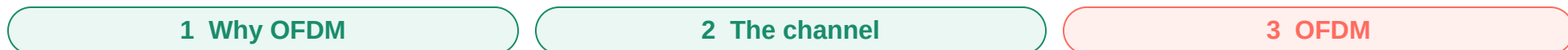
ONE 20 MHz 802.11ax SYMBOL, SPLIT BETWEEN FOUR USERS



a 242-tone channel divided into resource units, one per user

resource unit sizes	26 · 52 · 106 · 242 · 484 · 996 · 2×996 tones
smallest unit	26 tones, about 2 MHz
users in one 20 MHz symbol	up to 9
the LTE equivalent	a resource block of 12 subcarriers

OFDMA is OFDM plus a rule for who gets which subcarriers.



What OFDM fixed, and what it cost

WHAT OFDM FIXED

- ✓ the channel is flat on each subcarrier
- ✓ equalisation is one division, not a filter
- ✓ the prefix absorbs all the intersymbol interference
- ✓ wider channels cost nothing extra to equalise
- ✓ subcarriers can be handed to different users

WHAT OFDM COST

- ✗ 5 to 20% of the air time goes to the prefix
- ✗ the peak-to-average ratio is about 11 dB
- ✗ a frequency error breaks orthogonality
- ✗ notches still kill individual subcarriers
- ✗ nothing here made the link carry more streams

Every subcarrier is now a clean, flat, single-number channel, and one division corrects it.

Every subcarrier still carries only $\log_2(1 + \text{SNR})$ bits.

1 Why OFDM

2 The channel

3 OFDM